NELSON MANDELA UNIVERSITY

Faculty of Business and Economic Sciences

Department of Business Management
South Campus and George Campus

Financial Management
EBMV301
Previous codes: EBM301 and EBM351

Study Guide 2026

Compiled by:
Prof C Rootman
Prof J Krüger

Contributions by:
Dr S Watson
Ms E Kobese

Lecturers:

Prof C Rootman (module coordinator)	*South Campus*
Dr S Watson	*South Campus*
Ms E Kobese	*South Campus*
Ms N Maliwa	*George Campus*

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WORD OF WELCOME

Dear student

Welcome to EBMV301/EBM301/EBM351! May this be the best module of your entire university career! Know that the EBMV301/EBM301/EBM351 team is there to support you every step of the way.

CAREFULLY READ THE COMPLETE STUDY GUIDE! 😊

EBMV301 lectures will take place in-person (face-to-face lectures) in venues on the South and George Campuses during the 1st semester of 2026. Since Nelson Mandela University is a contact higher education institution and as per this module's offering by the Department of Business Management, it is COMPULSORY to attend these lectures.

Student lecture **slides** are available BEFORE each lecture, for you to use for note-taking. These slides (notes) are an additional resource, and it will NOT be sufficient to ONLY study the slides. **The lectures are the MAIN resource to use for this module** (all explanations, discussions, guidelines, etc. will be given DURING lectures), thereafter the slides, SG, tutorials, etc. are additional resources.

Remember, throughout this module, following announcements during lectures, the **Funda** site and **email** will also be key to our communication. Check, visit and respond/"click read receipts" (when necessary/requested) to communications via these means regularly!

This is a module at third-year level: Make sure that you **spend adequate time** on it, **engaging with the content and learning material** continuously. Also make sure that you participate in (submit/complete) all necessary assessments timely (formative and/or summative; online and/or in-person).

Also note that this module is NOT a Continuous Assessment (CA) module. Rather, the **module is an EXAMINABLE module**. This means that there are both formative (e.g. assignment and tutorials) and summative assessments (semester tests) that take place during the semester that accumulates to your class mark. In addition, there is a formal examination at the end of the module during the June 2026 examination period (as per the university schedule), that will have the highest weighting towards your final mark.

Make sure that you DO NOT MISS out on any of the assessments, which are all OPPORTUNITIES for you to build/improve your class mark before the examination. Then, with a strong performance during the examination, you can ensure a positive final mark and result.

It is important to note that we **do NOT** offer a summer/winter school for EBMV301/EBM301/EBM351. Therefore, you need to make sure that you work hard, attend your lectures, submit/complete all assessments and ask questions when in doubt. In doing so, you will increase your chances of being successful.

Please do not hesitate to contact anyone of us should you have any questions or concerns regarding the module. You will find our contact details in the study guide as well as on the first set of slides for this module.

Prof C Rootman (module coordinator)	*South Campus*
Dr S Watson	*South Campus*
Ms E Kobese	*South Campus*
Ms N Maliwa	*George Campus*

LECTURERS' DETAILS

Prof C Rootman (module coordinator) *South Campus*
Office: South campus, Main building, 11th Floor, Room 08
Direct telephone number: 041 504 4063
Email: chantal.rootman@mandela.ac.za

Dr S Watson *South Campus*
Office: South campus, Main building, 11th Floor, Room 10
Direct telephone number: 041 504 4065
Email: storm.watson@mandela.ac.za

Ms E Kobese *South Campus*
Office: South campus, Main building, 11th Floor, Room 20
Direct telephone number: 041 504 2745
Email: emihle.kobese@mandela.ac.za

Ms N Maliwa *George Campus*
Office: Admin building 0149
Direct telephone number: 044 801-5587
Email: nomatamsanqa.maliwa@mandela.ac.za
Email of the Administrative assistant on George Campus: charlene.cupido@mandela.ac.za

Consultation hours of lecturers: see details pasted on our respective office doors. You are also welcome to contact us via email to confirm an appointment time.

Departmental secretary: Ms N Adams *South Campus*
Office: Main building, 11th Floor, Room 04
Direct telephone number: 041 504 2201
Email: nasreen.adams@mandela.ac.za

Administrative assistant: Ms V Vinqi *South Campus*
Office: Main building, 11th Floor, Room 06
Direct telephone number: 041 504 4745
Email: vuyokazi.vinqi@mandela.ac.za

PRESCRIBED TEXTBOOK

Marx, J., De Swardt, C., Pretorius, M., Rosslyn-Smith, W. & Morake, M.B. 2023. *Financial management in Southern Africa*. 6th edition. Cape Town: Maskew Miller Learning.

This book is also available on short-loan via the NMU library.

CALCULATOR

You need a financial calculator such as the SHARP EL-738F or similar financial or business calculator (eg Casio, HP).

WORK SCHEDULE / PROGRAMME 2026

(As at January 2026 – schedule is subject to change if changes are made to the university calendar; schedule changes to be communicated via lectures, Funda and email)

Date	Day	Topic	Chapter	Presenter
9 Feb	**Monday**	**Introduction to the module**		**CR**
11 Feb	Wednesday	The financial goals of a firm	1	EK
16 Feb	Monday	The financial goals of a firm	1	EK
18 Feb	Wednesday	Understanding financial statements	3	SW
23 Feb	Monday	Analysing financial statements **Individual assignment opens online**	4	CR
25 Feb	Wednesday	Business and financial planning	5	CR
2 March	Monday	Business and financial planning	5	CR
4 March	Wednesday	Risk and return **Individual tutorial A opens online**	6	SW
9 March	Monday	Risk and return	6	SW
11 March	Wednesday	Time value of money **DUE DATE: Individual tutorial A**	7	EK
16 March	Monday	Valuation of shares and debentures	8	CR
18 March	Wednesday	Valuation of shares and debentures	8	CR
Between 18 and 20 March (TBC)	**Wednesday/ Thursday/ Friday (TBC)**	**Semester test 1 (Chapters 1, 3, 4, 5, 6 and 7)** **In-person at a university campus venue; date, time and venue to be confirmed**		
23 March	Monday	Net working capital and cash flow management	9	CR
25 March	Wednesday	Net working capital and cash flow management **DUE DATE: Individual assignment**	9	CR
RECESS Thursday 26 March – Monday 6 April				
8 April	Wednesday	The management of accounts receivable	10	SW
13 April	Monday	The management of inventory **Individual tutorial B opens online**	11	EK
15 April	Wednesday	Capital budgeting and cash flow principles	12	SW

Date	Day	Topic	Chapter	Presenter
20 April	Monday	Capital budgeting techniques **DUE DATE: Individual tutorial B**	13	CR
22 April	Wednesday	Capital budgeting techniques	13	CR
Between 22 and 23 April (TBC)	**Wednesday/ Thursday (TBC)**	**Semester test 2 (Chapters 8, 9, 10, 11 and 12)** **In-person at a university campus venue; date, time and venue to be confirmed**		
27 April	Monday	No lecture (Public holiday)		
29 April	Wednesday	Risk adjustments and other refinements to capital budgeting	14	SW
4 May	Monday	Risk adjustments and other refinements to capital budgeting AND revision	14	SW
6 May	Wednesday	The cost of capital	15	CR
11 May	Monday	The cost of capital and Dividend policy	15 & 18	CR
13 May	Wednesday	Leverage and capital structure	16	EK
18 May	Monday	Leverage and capital structure	16	EK
20 May	Wednesday	Dividend policy and **Module overview**	18 & review	CR
Assessment/Examination period commences Tuesday 26 May				
Re-assessment/Re-examination/Supplementary examinations (including last outstanding module examinations) – if applicable – take place during the July re-assessment period (6 – 10 July 2026), as determined by the university schedule (see policies and rules in the NMU Prospectus 2026).				

TBC: To be confirmed by the Timetabling Office

Note that this module equals 24 credits and is presented at NQF level 7.

CONTACT SESSIONS (LECTURES) AS PER INTRANET

South Campus

Period	Time	Mon	Wed
7	11:45 – 12:20	**L** 35 0 0040	**L** 35 0 0040
8	12:20 – 12:55	**L** 35 0 0040	**L** 35 0 0040

George Campus

Period	Time	Mon	Wed
11	14:25 – 15:00	**L** 301 1 0101	**L** 301 1 0101
12	15:00 – 15:35	**L** 301 1 0101	**L** 301 1 0101

You have to attend **TWO double lectures per week**. The first will take place on a Monday. The second will take place on a Wednesday. **PLEASE NOTE THAT THERE ARE NO EVENING LECTURES**. Check the work programme to ensure that you do not miss a lecture. Do not miss important module information or content that are shared during lectures. Do not miss out on a lecture and then try to "catch up" later in the semester. The amount of work for this module will not allow you to do this and end successfully… CONSISTENCY IS KEY – work hard throughout the semester.

Regarding "**class participation"**: You should be able to critically evaluate the contents of each study unit beforehand and after each session so that you are able to understand Financial Management concepts, examples, equations and calculations – critical for when you have to submit/complete assessments. Ask questions during the lectures or via email. The application of Financial Management concepts and calculations forms part of the development of important skills and is part of the learning process. **Remember, ATTENDANCE of EBMV301 LECTURES is COMPULSORY.**

In addition to this study guide, also refer to the Department of Business Management's general **information booklet** 2026. It is important to take note of the Department's policy regarding: Evaluations/assessments and illness/organised sport during such evaluations/assessments, amongst all the other policies and rules. Information (including rules, etc.) published in the **NMU Prospectus 2026** and those communicated via lectures and Funda/email takes president.

You are required to attend and work through ALL lectures as well as complete ALL assessments. You need to work hard and submit all your assessments on time in order for you to succeed. The pace of this module is fast and the content extensive. Your attitude should also reflect high levels of dedication and a real quest to increase your knowledge and application of financial management.

DETAILS OF ASSIGNMENT

More details on the assignment to follow during lectures and via Funda and/or email messages.

You have to submit **ONE individual** assignment (AS1) for this module. No late assignments will be accepted, since the online assignment quiz will already open on 23 February 2026 and close at 20:00 on the due date.

Due date for AS1: Wednesday, 27 March 2026. The cut-off time is 17:00.

The assignment will take the form of an **online quiz**. This is to be completed on the module's Funda site by **each individual student**. No hard copy assignments will be accepted.

This COMPULSORY assignment forms part of your class mark (15%) and is a formative assessment (open for a long period), and therefore there is **NO re-assessment (2nd assignment opportunity). If you fail to submit the assignment before the closing time on the due date, you will receive zero for this assessment.**

DETAILS OF TUTORIALS

You also have to complete **TWO compulsory** and **INDIVIDUAL** tutorials for this module. The process of accessing and completing tutorials will be explained in the first lecture. Note that tutorials are completed and then submitted online (via Funda) and no physical/in-person face-to-face tutorials are applicable for this module. No hard copy tutorials will be accepted. No late tutorials will be accepted, since the online tutorials quizzes will open in advance and close at 20:00 on the due dates. More details on the tutorials to follow during lectures and via Funda and/or email messages.

The **two** tutorials will be available online **on Funda in the form of multiple choice quizzes**. These are to be completed and submitted on the module's Funda site by **each individual student**. Each student's average mark of the tutorials will reflect as TT1 on ITS for this module.

Tutorial A questions will already open on 4 March 2026 and close at 17:00 on the due date. Tutorial B questions will already open on 13 April 2026 and close at 17:00 on the due date.

Due date for Tutorial A: Wednesday, 11 March 2026. The cut off-time is 17:00.
Due date for Tutorial B: Monday, 20 April 2026. The cut off-time is 17:00.

Tutorials are COMPULSORY and are formative assessments that form part of your class mark (20%). Without sufficient tutorial completion you will **NOT be able to "build" a sufficient class mark** (that will be valuable for the examination period and your final mark).

NOTE: There are NO tutorial re-assessments/duplicates. **These tutorials are formative assessments (open for a long period)** and therefore there is **NO re-assessment (2nd tutorial opportunities for each of these). If you fail to complete tutorials before the closing time on the due dates, you will receive zero for these missed tutorials.**

End-of-chapter problems in the prescribed textbook, exercises in the study guide and/or additional questions posted on Funda will be valuable for revision purposes for the tutorials – you can expect similar questions. The tutorial solutions will be available on Funda, after you have completed the tutorials and the marks have been published.

Should you have tutorial related questions and need assistance, please consult us during our consultation hours or e-mail us to make an appointment.

CALCULATION OF CLASS AND FINAL MARKS

Final mark = 50% class mark + 50% exam mark

The class mark consists of three components:

Average of the two semester tests (TM 1 and TM2)	65%
Assignment (AS1)	15%
Average of the two tutorials (TT1)	20%
Total	**100%**

NB: Both semester tests (summative assessments) are **compulsory**. **The average mark of the two will be used in the calculation of your class mark.** If only one test is written AND a **valid** doctor's certificate was submitted within three days after the missed test, an oral test may be given. No written sick test will be scheduled at this stage. NO marks will be doubled for tests missed for ANY reason. **Please note that you should have seen the doctor before or on the day of the test. NO doctor's certificate will be accepted if the doctor was seen AFTER the date of the test.** If an **invalid doctor's/hospital**

certificate/note was submitted, or no doctor's/hospital certificate/note was submitted in time (within three working days from missing the test), then an assessment mark of zero will be allocated to you for the missed assessment **(and then no "sick test"/assessment will be scheduled).**

Following the submission of a valid doctor's certificate, the arrangement of how this missed summative assessment (semester test) will then be managed, will be communicated to you. IF a "sick test"/assessment is to be written in place of the missed semester test, this "sick test"/assessment will be given **towards the END of the semester** and will cover **ALL** the work covered in **BOTH the originally scheduled** semester tests.

There is **no minimum class mark/Due Performance (DP) requirement** for admission to this module's examination. However, as mentioned above, your class mark is a combination of the average semester test mark, the assignment mark and the average tutorial mark. **Therefore, if no assignment was submitted and or if you did not complete/submit the tutorials, it implies that you will not have a high chance of passing the module**. In addition, the **semester test marks will add value to your final mark**, before the examination period. NO marks will be "doubled-up" for assessments missed for ANY reason.

Reminder: As indicated earlier, the assignment and tutorials are formative assessments and therefore there are no re-assessments for any missed assignment and tutorials. If you miss one of these assessments, you will be allocated a mark of zero for the missed assessment. Therefore, for every missed assessment (without a valid reason and supporting documentation), you will receive a mark of zero for that missed assessment.

As mentioned, no minimum class mark/(DP mark is required to be admitted to the examination at the end of the semester (June). However, a sub-minimum applies to the examination mark: you need a **minimum of 40% for the examination** in order to pass this module (irrespective of your class mark or final mark). A **minimum final mark of 50%** is required to **pass** the module. Should your **final mark be between 45% and 49%, you will be eligible for a supplementary examination in July 2026**. Should EBMV301 be your **last outstanding module following the June 2026 examinations, you will be eligible for a supplementary examination in July 2026** (see the relevant rules in the NMU Prospectus 2026). Note: Last outstanding modules of 1st semester modules (such as EBMV301) are NOT written later than July of the current year or in January of the following year. **Thus, once final marks have been calculated in July 2026, there are NO further re-examinations, etc. for students that were registered for EBMV301 during the 2026 academic year.** Note all supplementary examination and last outstanding module rules in the NMU Prospectus 2026.

Always remember that marks on Funda and ITS are not final until all processes, including internal and external moderation, have taken place.

FUNDA SITE

Important communiqués and general information will be placed on the EBMV301 Funda site for all students. The Funda enrolment procedure will be communicated via email (and also mentioned during the lectures).

In addition to the use of Funda, e-mails (also individual emails if necessary) will be sent if and when required. Therefore, please ensure that you also read emails sent to your official university email address (student email address). Kindly adhere to email etiquette when communicating or sending messages (e.g. address your recipient appropriately; share your name and contact details as part of your email end-note/greeting).

Information on lectures, semester test information as well as information on assignments and tutorials will be communicated during lectures, thereafter summary notices will be placed on Funda (and if necessary, communicated via email).

PERFORMANCE OBJECTIVES

OVERALL MODULE OBJECTIVE

The main goal of this study guide, which comprises a number of parts with different chapters, is to introduce you to Financial Management in a systematic manner. In this study guide we do not aim to emphasise only the practical application of Financial Management. Our main point of departure is the **fundamental financial theories**. With this purpose in mind, and also to motivate you, the financial theory is, however, with the aid of assignments and exercises, continuously related to business practice.

The overall course objective is to acquire the applicable **knowledge, understanding** and **application possibilities** of contemporary views with regard to fundamental financial management problems, such as investment, financing and dividend decisions, where financial decision-making is evaluated according to the criterion of maximum shareholder wealth.

GENERAL OBJECTIVES

The main achievements expected of students in the different chapters can be divided into three types according to performance area:

Firstly, you must always understand certain basic financial concepts, guidelines, principles, structures etc. and memorise them in such a way that you can reproduce them during a written assessment (i.e. tests and examinations). This field of study is known as *verbal information*. As a rule, relatively more achievements in the field of verbal information are expected of you at the beginning of each chapter than towards the end. The achievements expected of you in this field of study are usually specified for you as certain essential facts which you must be able to reproduce by for example *explaining, naming, listing and describing* them.

Secondly, in the field of intellectual expertise (proficiency, skill) and comprehension, certain elaborations, interpretations, applications and the identification of mutual relationships within the financial subject area will be expected of you. You will, for example, be required to relate new data or observations to acknowledged financial theories to find solutions to practical problems by applying certain guidelines which have been learnt verbally. Typical activities to sharpen and test your skill in this field of study are multiple choice questions, classification exercises, completion of worksheets and particularly the end-of-chapter problems in the textbook, study guide and on Funda.

Thirdly, it is an important endeavour of all your lecturers in Business Management to imbue every student who takes his/her studies seriously with a positive attitude towards the subject. Because this objective is pursued throughout, it is not repeated in the wording of the specific objectives which you will receive with regard to each chapter. You will be able to demonstrate your disposition by the standard and perseverance of your achievement, your genuine interest in Financial Management, participation in class, the questions you ask, the respect you show for the language, feelings, etc. of others, your conversations with your lecturer and your willingness to do more than the prescribed minimum that is expected of you.

LEARNING OUTCOMES

With regard to the theme of each chapter, you should be capable of:

- reproducing the essential content by naming and/or explaining the relevant content (where possible with the aid of diagrammatic representations);
- identifying the meaning reflected in the content by indicating the most acceptable or correct possibility from various given alternatives (e.g. by answering true/false questions, multiple choice questions, etc.);
- demonstrating your ability to establish relationships, e.g. between theory and practice, *inter alia*, by completing worksheets and generating solutions to problems; and
- generating solutions for any relevant extensive problems by completing work assignments, case studies or critical analyses.

Each of the learning outcomes is elaborated on and some examples provided where applicable.

OBJECTIVE-ORIENTATED STUDY METHOD

The following study method is recommended in the module:

- Attentively read through the learning outcomes of the particular chapter before attending the respective lecture(s) and/or attempting the assignments as well as tutorials. The assignment and tutorials should only be attempted after you attended the applicable lectures and after fully covering the relevant chapters’ content.
- Study the appropriate chapter and respond to the learning outcomes of each chapter.
- The lecture slides are only additional study tools.
- Work through the applicable end-of-chapter overviews, reviews, self-test questions and critical-thinking exercises.
- Use the study guide question examples for revision purposes.
- Use the lecture discussions (in class) and lecturer guidelines while you study for any of the assessments.
- Discuss any problems with your lecturers timeously.

PART ONE: FUNDAMENTALS OF FINANCIAL MANAGEMENT

CHAPTER 1: THE FINANCIAL GOALS OF A FIRM

INTRODUCTION

This chapter introduces some basic ideas in corporate finance as linked to the financial goals of a firm. The chapter covers strategies, business forms, long-term financial goals, short-term financial goals, functions of a financial manager and three fundamental financial principles.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- describe the reasons for the existence of business firms
- explain the need for a vision, mission and organisational culture of a firm
- explain the role of the business model and strategy in managing a firm
- evaluate the various forms of business organisation
- distinguish between the long-term and short-term financial goals of a firm
- relate the functions of a financial manager to the long- and short-term financial goals of a firm
- describe the competencies of a chief financial officer (CFO)
- apply the fundamental principles of financial management.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. **TRUE OR FALSE:** Firms exist to only satisfy consumers' needs by providing products and services.

(a) True
(b) False

2. The following statement is **CORRECT**:

(a) A mission statement reflects what a firm aims to look like in the future.
(b) A vision statement reflects a firm's current reason for existence and for the future.
(c) Sustainability refers to a firm's ability to endure and ensure that its actions are environmentally bearable, socially equitable and economically (financially) viable.
(d) Most experts believe that business strategy is more important than organisational culture.
(e) Economists refer to a recession when the economic growth declined for four consecutive quarters.

3. The possibility of conflict of interest between the shareholders and management of the firm is called

(a) the shareholders' conundrum.
(b) corporate breakdown.
(c) the agency problem.
(d) the agency relationship.
(e) legal liability.

4. The cost of capital of a firm includes the following:

(a) cost of debt.
(b) cost of debt, cost of inventory, cost of assets.
(c) cost of debt, cost of preference shares, cost of assets.
(d) cost of debt, cost of ordinary shares.
(e) cost of debt, cost of preference shares, cost of ordinary shares.

5. A financial manager's main responsibilities can be grouped into the following areas:

(a) investments, risk and return, financing.
(b) working capital management, investments, financing.
(c) working capital management, short-term investments, long-term investments.
(d) short-term financing, long-term financing, inventory.
(e) inventory, cash, accounts receivable.

Answers to short questions

1. b
2. c
3. c
4. e
5. b

EXAMPLES OF OPEN QUESTIONS

1. Google PEPKOR Holdings Limited and consider their PEP Stores brand. Then answer the following questions:

(a) Which strategy do you think PEP Stores is following? Motivate your answer by discussing and by applying the strategy to PEP Stores.
(b) Which form of business does PEPKOR Holdings Limited operate under? Explain your answer.
(c) By considering your answer to (b) above, discuss one 'advantage' of this form of business to the owners of the firm.
(d) By considering your answer to (b) above, discuss one requirement in terms of the financial statements of this form of business that needs to be adhered to.

2. Consider the financial information of three companies and answer the questions that follow.

Ro Save Group Ltd

	Mar 25 Interim	Sept 24 Final	Sept 23 Final	Sept 22 Final	Sept 21 Final
Attributable income (Rm)	520	524	953	916	745
Return on shareholders' interest (%)	38.50	40.93	40.70	44.40	43.50
Price Period End (c)	10 825	11 575	9 629	9 290	6 470

Buy Now Stores Ltd

	Feb 25 Interim	Feb 24 Final	Feb 23 Final	Feb 22 Final	Feb 21 Final
Attributable income (Rm)	1 050	1 114	785	1 189	1 055
Return on shareholders' interest (%)	23.80	29.90	36.50	52.70	65.10
Price Period End (c)	4 520	4 307	4 644	4 040	3 100

Choice Ltd

	Jun 25 Interim	Jun 24 Final	Jun 23 Final	Jun 22 Final	Jun 21 Final
Attributable income (Rm)	3 250	3 027	2 510	2 267	1 998
Return on shareholders' interest (%)	22.58	23.76	35.41	38.30	40.14
Price Period End (c)	16 550	15 067	10 180	8 285	5 500

(a) Which company ranks first in terms of profit maximisation for the year 2025? Motivate your answer.
(b) Which company ranks first in terms of rate of return maximisation for the year 2025? Motivate your answer.
(c) Which company ranks first in terms of the maximisation of shareholders' wealth (i.e. share price maximisation) over the last 12-month period? Motivate your answer by also including calculations.

Solution

(a) Choice Ltd, since it has the highest attributable income (or net profit or earnings after tax) for 2025 of R 3 250 million.
(b) Ro Save Group Ltd, since it has the highest return on shareholders' interest (or return on equity) for 2025 of 38.50%.
(c) Choice Ltd, since it has the highest positive change in share price over the 12-month period. The increase in share price is 9.84%:

% change in share price = [(New price – Old price) / Old] x 100

	P_t	P_{t-1}	$(P_t - P_{t-1})/P_{t-1}$ x 100
Ro Save Group Ltd	10825c	11575c	-6.48%
Buy Now Stores Ltd	4520c	4307c	4.95%
Choice Ltd	16550c	15067c	9.84%

PART ONE: FUNDAMENTALS OF FINANCIAL MANAGEMENT

CHAPTER 3: UNDERSTANDING FINANCIAL STATEMENTS

INTRODUCTION

This chapter covers the topic of financial statements by considering the users of financial statements and the accepted accounting standards and principles (IFRS and GAAP). The classification of financial information is addressed. The various financial statements summarising the financial information are also discussed. The information contained in the auditors' and directors' reports are also presented.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- identify the users of financial statements
- briefly describe the International Financial Reporting Standards
- explain how the types of accounts may be classified
- record changes in the financial position
- summarise financial information in the financial statements
- interpret the auditors' report and the directors' report.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. The statement of _________ provides a summary of how the earnings after tax were determined for a particular accounting period.

(a) financial position
(b) financial performance
(c) retained earnings
(d) cash flow
(e) financial reporting

2. All of the following are Generally Accepted Accounting Principles, **EXCEPT**:

(a) money conservatism.
(b) going concern.
(c) accounting entity.
(d) market cost.
(e) double entry.

3. Which one of the following is *incorrect* in terms of the accounting period?

(a) Month
(b) Quarter of a year
(c) Semi-annually (half a year)
(d) Three-quarters of a year
(e) Year

4. The statement of financial position identity (accounting equation) states that

(a) current assets + non-current assets = total assets
(b) assets = liabilities + owners' equity
(c) current liabilities + non-current liabilities = total liabilities
(d) ordinary shares + retained earnings = shareholders' equity
(e) cash flow = market value – book value

5. The _________ provides an overview of the firm and its state of affairs.

(a) auditors' report
(b) governance report
(c) sustainability report
(d) directors' report
(e) cash flow statement

Answers to short questions

1. b
2. d
3. d
4. b
5. d

EXAMPLES OF OPEN QUESTIONS

1. Why is depreciation classified as a non-cash charge?

Solution

Depreciation is a non-cash charge shown as an expense in the statement of financial performance that do not actually involve an actual cash outflow. Therefore, depreciation does not represent an actual cash outflow.

2. Referring to PEPKOR Holdings Limited (see Examples of open questions for Chapter One). View the Annual Financial Statements and answer the following questions:

(a) Identify the applicable accounting period of the latest Annual Financial Statements.
(b) Consider the Directors' report and identify whether there was any movement in ordinary shares.
(c) What debt did PEPKOR obtain during the financial year when considering the Directors' report?

Solution

(a) 1 October 2024 to 30 September 2025

(b) Issued 9.8 million (2024: 17.9 million) ordinary shares during the current financial year for share rights that vested under the Pepkor Executive Share Rights Scheme.
The group did not repurchase or cancel ordinary shares during the current financial year from the open market on the Johannesburg Stock Exchange (JSE). In the prior year, the group repurchased and cancelled 1.7 million ordinary shares from the open market on the JSE.
A company within the group transferred 508 175 (2024: 443 095) ordinary shares for share rights that vested under the Pepkor Scheme.

(c) The group successfully raised R2.083 billion in the bond market following an auction held on 5 March 2025. The group further successfully placed an additional R1.250 billion under its Domestic Medium Term Note programme, raised R3.5 billion in term loans and entered into a new R1.0 billion revolving credit facility, effective from the end of March 2025.
The group also entered into a new R2.0 billion revolving credit facility, effective from December 2024. The proceeds from the above replaced R206 million in floating rate notes, R5.205 billion in term loans and R2.5 billion in revolving credit facilities.

PART ONE: FUNDAMENTALS OF FINANCIAL MANAGEMENT

CHAPTER 4: ANALYSING OF FINANCIAL STATEMENTS

INTRODUCTION

This chapter focuses on how to analyse financial statements. Analysing financial statements are analysed by doing industry comparative and time-series analyses and calculating financial ratios. Several considerations when calculating and interpreting financial ratios are also highlighted.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- distinguish between industry comparative and time-series analysis
- describe the considerations to be kept in mind when performing ratio analysis
- accurately calculate profitability, liquidity, activity, solvency and securities market ratios
- evaluate each ratio and suggest possible corrective actions (if necessary).

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. Earnings per share (EPS) is equal to

(a) net profit after tax divided by the total number of shares outstanding or in issue.
(b) Net income divided by the par value of ordinary shares.
(c) gross income multiplied by the par value of ordinary shares.
(d) operating income divided by the par value of ordinary shares.
(e) Net income divided by total shareholders’ equity.

2. Dividends per share (DPS) is equal to dividends paid

(a) divided by the par value of ordinary shares.
(b) divided by the total number of shares outstanding or in issue.
(c) divided by total shareholders’ equity.
(d) multiplied by the par value of ordinary shares.
(e) multiplied by the total number of shares outstanding or in issue.

3. __________ is the difference between a firm's current assets and its current liabilities.

(a) Operating cash flow
(b) Capital spending
(c) Net working capital
(d) Cash flow from assets
(e) Cash flow to creditors

4. The financial ratio measured as the price per share divided by earnings per share (EPS) is known as the

(a) return on assets.
(b) return on equity.
(c) debt-equity ratio.
(d) price-earnings ratio.
(e) Du Pont identity.

5. Ratios that measure how efficiently a firm uses its assets to generate sales are known as __________ ratios.

(a) asset management
(b) long-term solvency
(c) short-term solvency
(d) profitability
(e) market value

Answers to short questions

1. a
2. b
3. c
4. c
5. a

EXAMPLES OF OPEN QUESTIONS

1. Assume the Furniture Group Ltd was founded in 1934 and listed on the JSE in 2004. This company is a leading retailer of household furniture, electrical appliances and home electronics sold on credit mainly through various well-known brands. The company is classified as follows:

- Industry: Consumer services
- Super-sector: Retail
- Sector: General retailers
- Sub-sector: Home improvements

Consider the ratios below and perform a trend and a comparative analysis of the Furniture Group Ltd.

	Mar 25 Final	Mar 24 Final	Mar 23 Final	Mar 22 Final	Mar 21 Final
ROE (%)	18.73	19.10	18.07	19.35	23.53
ROA (%)	19.05	18.71	17.77	19.89	24.82
Operating profit margin (%)	23.45	23.00	22.07	21.86	25.87
D:E	0.29	0.33	0.34	0.29	0.28
Current ratio	4.65	3.40	3.53	3.23	3.08
Dividend cover	2.05	2.15	2.08	1.97	2.22

Assume that the average ratios in the “General Retailers sector” are as given below. Perform a comparative analysis of Furniture Group Ltd versus the sector.

	2024	2023	2022	2021
ROE (%)	23.43	23.62	20.25	18.18
ROA (%)	21.45	21.60	19.44	17.84
Operating profit margin (%)	21.01	21.04	20.57	19.10
D:E	1.25	1.21	1.18	1.07
Current ratio	3.63	3.67	3.81	3.74
Dividend cover	2.10	2.46	2.05	2.28

Solution

1.
ROE
- Interpretation 2025 figure: For every R1 shareholders invested in the firm, they received 18.73c in return.
- This ratio decreased over time if considered as from 2021 but increased from 2023 to 2024. However, there was a slight decline from 2024 to 2024.
- In 2022, the firm’s ROE was above the average of the general retailers sector, but as from 2023 to 2025 it was below the average.

ROA
- Interpretation 2025 figure: For every R1 lenders and shareholders invested in the firm, they received 19.05c in return.
- This ratio decreased considerable from 2021 to 2023, but increased from 2023 to 2025.
- In 2022, the firm’s ROA was above the average of the general retailers sector, but as from 2023 to 2025 it was well below the average.

Operating profit margin
- Interpretation 2025 figure: For every R1 made in sales, an operating profit of 23.45c was made.
- There has been a slight decrease in this margin from 2021 to 2022 but increased as from 2022 to 2025.
- For the period 2021 to 2025 the Furniture Group Ltd had a higher (better) operating profit margin compared to others in the same industry.

D/E
- Interpretation 2025 figure: For every R1 of Total Equity, the Furniture Group has R0.29 in Total Debt

- They have consistently increased the amount of debt they have relative to equity from 2021 to 2023, and a slightly decreased it from 2023 to 2025.
- The Furniture Group Ltd had considerably less debt compared to the industry average for the period 2021 to 2025. The industry has also consistently increased their usage of debt during this period.

Current ratio

- Interpretation 2025 figure: For every R1 in CL, the Furniture Group has R4.65 in CA.
- Ratio consistently greater than 1 and for the last five years it was higher than the standard norm of 2.
- As from 2021 to 2022, the current ratio was slightly lower than the industry average, but in 2025 the current ratio was higher than that of the industry.

Dividend cover

- Interpretation 2025 figure: A dividend cover of 2.05 implies that the Furniture Group has sufficient earnings to pay dividends amounting to 2.05 times of the present dividend payout during the period.
- A dividend cover of 2.05 implies that the Furniture Group has sufficient earnings to pay dividends amounting to 2.05 times of the present dividend payout during the period.
- Dividend cover changed over the last few years as some up and down movements can be noticed. It decreased from 2021 to 2022, increased from 2023 to 2024 with a slight decline in the ratio for 2025. For the period 2021 to 2025, the dividend cover was lower than the industry average, except for 2023 when the dividend cover was slightly higher than the industry average.

2. Given a net profit margin = 15%, ROE = 25%, D/E = 1.25, and assets = R800, calculate sales.

Solution

D/E = 1.25 (For every R1 in equity, there is R1.25 in debt, thus total capital = 2.25) this implies that

Equity	= 1/2.25 = 44.44% (Reworking of D/E ratio)
If total assets	= R800, then total capital is also = R800
Equity is thus:	0.4444 x 800 = R355.56

ROE	=	Net income / Equity
0.25	=	Net income / 355.56
Net income	=	R88.89

Net profit margin	=	Net income / Sales
0.15	=	88.89 / Sales
0.15 x Sales	=	88.89
Sales	=	88.89 / 0.15
Sales	=	R592.60

PART ONE: FUNDAMENTALS OF FINANCIAL MANAGEMENT

CHAPTER 5: BUSINESS AND FINANCIAL PLANNING

INTRODUCTION

This chapter considers elements relating to firms' business and financial planning. It covers risk assessment, operational planning and the budgeting process. With a future-orientation, the chapter focuses on compiling budgeted statements of financial performance and budgeted statements of financial position.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- link strategy, risk identification and assessment to financial planning
- compile a budgeted statement of financial performance
- compile a budgeted statement of financial position
- evaluate a budgeted statement of financial performance and a budgeted statement of financial position
- apply the principles of budgeting.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. All of the following statements regarding the percentage of sales method are correct, **EXCEPT**:

(a) all short-term debt will increase/decrease at the same rate as sales.
(b) costs will increase/decrease at the same rate as sales.
(c) accounts payable is a current liability will increase/decrease at the same rate as sales.
(d) the additional funds needed (AFN) amount can be positive or negative.
(e) a few rotations will be done for a particular year.

2. A firm's pay-out ratio refers to its

(a) percentage of earnings after tax that is retained as internal financing.
(b) ratio of total assets to sales.
(c) percentage of earnings after tax that is paid out as dividends.
(d) ratio of retained earnings to total assets.
(e) percentage of sales that is retained for internal financing.

3. A firm's retention ratio refers to its

(a) percentage of earnings after tax that is retained as internal financing.
(b) ratio of total assets to sales.
(c) percentage of earnings after tax that is paid out as dividends.
(d) ratio of retained earnings to total assets.
(e) percentage of sales that is retained for internal financing.

4. If a firm is at full-capacity sales, it means the firm is at the maximum possible level of production without increasing

(a) net working capital.
(b) cost of goods sold.
(c) inventory.
(d) non-current assets.
(e) the debt ratio.

5. A firm is currently operating at full capacity. Net working capital, costs, and all assets vary directly with sales. The firm does not wish to obtain any additional equity financing. The dividend pay-out ratio is constant at 40%. When the firm compiles a pro forma statement of financial position (balance sheet), the plug variable is most likely going to be

(a) accounts payable.
(b) long-term debt.
(c) non-current assets.
(d) retained profits.
(e) ordinary shares.

Answers to short questions

1. a
2. c
3. a
4. d
5. b

EXAMPLES OF OPEN QUESTIONS

1. Food Distributors Ltd uses the percentage of sales method as a financial planning model to perform long-term financial planning. Consider the firm's current statement of financial performance and statement of financial position and use the additional information to perform a percentage of sales method analysis by answering the questions that follow.

STATEMENT OF FINANCIAL PERFORMANCE OF FOOD DISTRIBUTORS LTD FOR THE YEAR ENDED 31 MARCH 2025 (R IN MILLIONS)

Sales	160 000
Costs of goods sold	90 000
Earnings before tax	70 000
Taxes (27%)	18 900
EARNINGS AFTER TAX	51 100

STATEMENT OF FINANCIAL POSITION OF FOOD DISTRIBUTORS LTD AS AT 31 MARCH 2025 (R IN MILLIONS)

ASSETS		**EQUITIES AND LIABILITIES**	
Non-current assets		Owner's equity	
Net plant and equipment	200 000	Ordinary share capital	180 000
		Retained earnings	30 000
Current assets		**Total equity**	210 000
Inventory	80 000		
Accounts receivables	55 000	Non-current liabilities	
Cash	25 000	Long-term loan	130 000
Total current assets	160 000		
		Current liabilities	
		Accounts payable	17 000
		Short-term debt	3 000
		Total current liabilities	20 000
TOTAL ASSETS	**360 000**	**TOTAL EQUITY AND LIABILITIES**	**360 000**

Additional information:

- Mainly due to the effects of loadshedding and watershedding disruptions, the firm has projected a small 5% **increase** in sales for the coming year.
- The firm has also predicted that costs, assets and accounts payable are proportional to sales.
- Equity, long-term and short-term debt are not proportional to sales.
- The tax rate will remain constant.
- For the financial year that ended March 2025, R30 660 of the earnings after tax (net income) was paid out as dividends. The firm's dividend pay-out ratio will remain the same.

(a) Complete the *budgeted* or *pro forma* statement of financial performance and *pro forma* statement of financial position for the firm.

BUDGETED/PRO FORMA STATEMENT OF FINANCIAL PERFORMANCE OF FOOD DISTRIBUTORS LTD FOR THE YEAR ENDED 31 MARCH 2026 (R IN MILLIONS)

Sales	
Costs of goods sold	
Earnings before tax	
Taxes (27%)	
EARNINGS AFTER TAX	

BUDGETED/PRO FORMA STATEMENT OF FINANCIAL POSITION OF FOOD DISTRIBUTORS LTD AS AT 31 MARCH 2026 (R IN MILLIONS)

ASSETS	
Non-current assets	
Net plant and equipment	
Current assets	
Inventory	
Accounts receivable	
Cash	
Total current assets	
TOTAL ASSETS	
EQUITY AND LIABILITIES	
Owner’s equity	
Ordinary share capital	
Retained earnings	
Total equity	
Non-current liabilities	
Long-term loan	
Current liabilities	
Accounts payable	
Short-term debt	
Total current liabilities	
TOTAL EQUITY AND LIABILITIES	

(b) Calculate the firm's additional funds needed (AFN).
(c) Discuss three possible sources of capital to ensure that the firm's statement of financial position balances.
(d) Calculate the firm's 2026 forecasted dividend pay-out (in Rands).
(e) What is the firm's retention ratio?

Solution

(a)

BUDGETED/PRO FORMA STATEMENT OF FINANCIAL PERFORMANCE OF FOOD DISTRIBUTORS LTD FOR THE YEAR ENDED 31 MARCH 2026 (R IN MILLIONS)

Sales	168 000
Costs of goods sold	94 500
Earnings before tax	73 500
Taxes (27%)	19 845
EARNINGS AFTER TAX	53 655

BUDGETED/PRO FORMA STATEMENT OF FINANCIAL POSITION OF FOOD DISTRIBUTORS LTD AS AT 31 MARCH 2026 (R IN MILLIONS)

ASSETS	
Non-current assets	
Net plant and equipment	210 000
Current assets	
Inventory	84 000
Accounts receivable	57 750
Cash	26 250
Total current assets	168 000
TOTAL ASSETS	378 000
EQUITY AND LIABILITIES	
Owner's equity	
Ordinary share capital	180 000
Retained earnings	51 462
Total equity	231 462
Non-current liabilities	
Long-term loan	130 000
Current liabilities	
Accounts payable	17 850
Short-term debt	3 000
Total current liabilities	20 850
TOTAL EQUITY AND LIABILITIES	382 312

Workings to calculate *budgeted/pro forma* Retained earnings (RE)
RE of 2025 + RE to occur in *budgeted/pro forma* year of 2026
= 30 000 + (0.40 x 53 655)
= 30 000 + 21 462
= R51 462

(b)

AFN	=	Total Assets – Total equity and liabilities
	=	R378 000 – R382 312
	=	-R4 312

(c)
Any three of the following:

Decrease / pay short-term debt
Decrease / pay long-term debt
Decrease share capital / share buy-back
Increase dividend pay-outs to decrease (reduce) retention ratio (retained earnings)

(d)
30 660 of 51 100 was paid out as dividends in 2025; thus, the dividend pay-out ratio is 60%

2026 Dividend	= 0.60 x Earnings after tax
	= 0.60 x 53 655
	= R32 193

OR
Can also work with retention ratio and amount: 2026 Dividend = 2026 Earnings after tax – Retained earnings 2026 addition which is 40% of the Earnings after tax
= 53 655 – 21 462
= R32 193

(e)
Dividend pay-out ratio is 60%, therefore retention ratio is 40%
(can also do calculation from 2025 figures; answer must just reflect as 40%)

2. Sundown Caravans Ltd used the percentage of sales method to perform its financial planning for 2026. Its budgeted/pro forma statement of financial position shows its Total assets as R2 500 600 and its Total equity and liabilities as R2 350 000.

(a) Calculate the firm's additional funds needed (AFN).
(b) Discuss four possible sources of capital to ensure that the firm's statement of financial position balances.

Solution

(a)

AFN	=	Total Assets – Total equity and liabilities
	=	R2 500 600 – R2 350 000
	=	R150 600

(b)
Increase short-term debt
Increase long-term debt
Increase share capital (by issuing/selling shares)
Decrease dividend pay-outs to increase (reduce) retention ratio (retained earnings)

3. When you realise that a firm is only operating at 70% capacity, will the initially calculated AFN increase or decrease when you do the *budgeted/pro forma* financial statements? Explain your answer.

4. Rent for a building is regarded as a fixed cost. Does this mean that a firm's rent bill can never increase? Explain your answer.

PART ONE: FUNDAMENTALS OF FINANCIAL MANAGEMENT

CHAPTER 6: RISK AND RETURN

INTRODUCTION

This chapter covers the annual and average multiple returns as well as risk. Risk is unpacked in terms of external and internal risks and how to measure risk. The risk and returns for both individual securities and portfolios are addressed. To assist with the determining of the risk and return of individual securities and portfolios, a discussion of the capital asset pricing model is presented. A brief discussion of the efficient market hypothesis is given.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- calculate rates of return
- identify and evaluate sources of risk
- measure risk by means of variance, standard deviation and coefficient of variation
- determine the return of a portfolio
- determine the required rate of return by means of the capital asset pricing model (CAPM)
- use arbitrage pricing theory (APT) in order to determine the required rate of return.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

Pay particular attention to the section on the slides dealing with the standard deviation for portfolios.

To calculate the expected return of a portfolio of assets, we use the following equation:

$$E(R_{portfolio}) = \sum_{j=1}^{n} w_j \ x \ E(R_j)$$

Where $E(R_{portfolio})$ = the expected return of a portfolio consisting of risky assets; n = the number of assets in the portfolio; w_j = weight of asset j in the portfolio; $E(R_j)$ = the average expected return of asset j.

Unfortunately calculating the standard deviation of a portfolio is not as straightforward as calculating its expected return, i.e. it is not just the weighted average of the individual assets' standard deviations. We need the correlation coefficient or the covariance. Once we've calculated the correlation coefficient between two securities (say A and B), we can calculate a portfolio's variance and standard deviation as follows:

$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B \sigma_A \sigma_B \rho_{AB}$ and $\sigma_p = \sqrt{\sigma_p^2}$

The lower the correlation, the better the lower the portfolio's standard deviation.

EXAMPLES OF SHORT QUESTIONS

1. The correlation coefficient between Share A and market is 0.8375. The standard deviations for Share A and the market portfolio are 1.85 and 1.25 respectively. Calculate Share A's volatility in relation to the riskiness of the market portfolio (beta coefficient).

(a) 0.36
(b) 0.99
(c) 1.05
(d) 1.24
(e) 2.31

2. What is the standard deviation of a portfolio that is invested 40% in share Q and 60% in share R? Assume that the correlation coefficient between the two shares is +1 (i.e. the shares are perfectly positively correlated).

State of Economy	Probability of State	Returns if State Occurs: Share Q	Returns if State Occurs: Share R
Boom	25%	18%	9%
Normal	75%	9%	5%

(a) 0.70 %
(b) 1.48 %
(c) 2.60 %
(d) 6.75 %
(e) 1.70 %

3. The risk-free rate of return is 4% and the market risk premium is 8%. What is the expected rate of return on a share with a beta of 1.28?

(a) 9.12%
(b) 10.24%
(c) 12.00%
(d) 13.12%
(e) 14.24%

4. You own 50 shares of Company A, which has a price of R12 per share, and 100 shares of Company B, which has a price of R3 per share. What is the portfolio weight of Company A's shares in your portfolio?

(a) 25%
(b) 33%
(c) 50%
(d) 67%
(e) 75%

5. Portfolio risk comprises of

(a) firm-specific risk and diversifiable risk.
(b) systematic risk minus unsystematic risk.
(c) diversifiable risk and unsystematic risk.
(d) market risk and firm-specific risk.
(e) market risk and non-diversifiable risk.

Answers to short questions

1. d
2. c
3. e
4. d
5. d

EXAMPLES OF OPEN QUESTIONS

1. Share A has the following probability distribution of forecasted returns:

Probability	Rate of return (%)
0.1	-15
0.2	0
0.4	5
0.2	10
0.1	25

Calculate share A's expected rate of return, variance and standard deviation. Show the relevant equations.

Solution

Average expected return $E(R_A) = \sum_{i=1}^{n} R_i \Pr_i$ =

= (-15)(0.1) + 0(0.2) + (5)(0.4) + (10)(0.2) + (25)(0.1) =5%

$$Variance = \sigma_A^2 = \sum_{i=1}^{n} \left(R_i - E\left(R_A\right)\right)^2 \; x \Pr_i$$

= $(-15 – 5)^2(0.1) + (0 – 5)^2(0.2) + (5 – 5)^2(0.4) + (10 – 5)^2(0.2) + (25 – 5)^2(0.1)$
= 40 + 5 + 0 + 5 + 40 = 90

$$Std\ dev = \sigma_A = \sqrt{\sigma_A^2}$$

= 9.49%

2. Use the following information to calculate the expected return and standard deviation for these two shares. Also calculate the coefficient of variation for both shares and indicate which one of the two a risk-averse investor would choose.

State of economy	**Probability of state of economy**	**Rate of return**	
		Share A	**Share B**
Recession	0.3	6%	-20%
Normal	0.6	7%	13%
Boom	0.1	11%	33%

Solution

The average expected return of a share is the sum of the probability of each return occurring times the probability of that return occurring. So, the average expected return of each share is

$$E(R_{Risky\ asset}) = \sum_{i=1}^{n} R_i \ \mathrm{Pr}_i$$

$E(R_A)$ = (0.06)0.30+ (0.07)0.60 + (0.11)0.10 = 0.0710 or **7.10%**
$E(R_B)$ = (-0.2)0.30 + (0.13)0.60 + (0.33)0.10 = 0.0510 or **5.10%**

To calculate the standard deviation, we first need to calculate the variance. To find the variance, we find the squared deviations from the expected return. We then multiply each possible squared deviation by its probability, then add all of these up. The result is the variance. So, the variance and standard deviation of each share is:

$$Variance = \sigma^2_{Risky\ asset} = \sum_{i=1}^{n} \mathrm{Pr}_i \ x \left(R_i - E(R_{Risky\ asset}\right)^2$$

σ_A^2 =0.30(0.06 – 0.0710)^2 + 0.60(0.07 – 0.0710)^2 + 0.10(0.11 – 0.0710)^2 = 0.000189
σ_B^2 =0.30(-0.20 – 0.0510)^2 + 0.60(0.13 – 0.0510)^2 + 0.10(0.33 – 0.0510)^2 = 0.03043

$$Std\ dev = \sigma_{Risky\ asset} = \sqrt{\sigma^2}$$

σ_A = (0.000189)$^{1/2}$ = 0.01375 or **1.38%**
σ_B = (0.03043)$^{1/2}$ = 0.1744 or **17.44%**

If you have to choose between the two shares, which one would you select? To answer this question, we need to calculate the Coefficient of Variation (CV):

CV = σ / E(R)
Share A: CV = 1.38 / 7.10 = **0.20**
Share B: CV = 17.44 / 5.10 = **3.42**

From the above, it is clear that Share A has less risk-per-unit of return (for every 1% of expected return, it only has 0.20% volatility as compared to Share B's 3.42% volatility).

3. You own a share portfolio comprising of a 25% investment in Share Q, 20% investment in Share R, 15% investment in Share S and 40% investment in Share T. The betas for these four shares are 0.65, 1.75, 1.25 and 1.54 respectively. Calculate and interpret the portfolio beta.

Solution

The beta of a portfolio is the sum of the weight of each asset times the beta of each asset. So, the beta of the portfolio is:

$$\beta_p = \sum_{j=1}^{n} w_j \beta_j$$

$\beta_p = 0.25(0.65) + 0.20(1.75) + 0.15(1.25) + 0.40(1.54) = \mathbf{1.32}$

Interpretation: The returns of this portfolio are slightly more reactive to changes in the overall stock market than the average portfolio i.e. if market returns, as measured by the FTSE/JSE ALSI, increase (or decrease) with 10% the returns of this portfolio will increase (or decrease) with 13.20%.

4. Suppose Ms Smith holds 100 SuperWare shares and 300 Intelligence shares. SuperWare shares are currently prices at R80 per share, while Intelligence shares are R40 per share. The expected return of SuperWare shares is 15% while that of Intelligence share is 20%. The standard deviation of SuperWare is 8% and that of Intelligence is 20%. The correlation coefficient between the shares is 0.29.

4.1 Calculate the expected return and standard deviation of the portfolio.

4.2 Today she sold 200 Intelligence shares to pay school fees. Calculate the expected return and standard deviation of her new portfolio.

Solution

4.1 We start by calculating the market value of the two holdings. Market value, also called market capitalisation, is calculated by multiplying the number of shares in issue with the current share price:

SuperWare: 100 x R80 = R8 000
Intelligent: 300 x R40 = R12 000
Total investment = R8 000 + R12 000 = R20 000
We then calculate the weights of each share in the portfolio
$w_{SuperWare}$: R 8 000 / R20 000 = 0.4
$w_{Intelligent}$: R12 000 / R20 000 = 0.6

The expected return and standard deviation are calculated as before, using the weighted average for the return and the variance of the portfolio using the formula for the risk of a two-asset portfolio:

$$E(R_p) = \sum_{j=1}^{2} w_j \ x \ E(R_j)$$

$R_P = 0.4(0.15) + 0.6(0.20) = 0.1800 = 18.00\%$

$$\sigma_p^2 = w_M^2\sigma_M^2 + w_I^2\sigma_I^2 + 2w_M w_I \sigma_M \sigma_I \rho_{MI} \quad \text{and} \quad \sigma_p = \sqrt{\sigma_p^2}$$

$\sigma_P^2 = 0.4^2(0.08)^2 + 0.6^2(0.2)^2 + 2(0.4)(0.6)(0.29)(0.08)(0.20) = 0.0176512$

$\sigma_P = (0.0176512)^{½} = 0.13286 = 13.29\%$

4.2 There is no change in the market value of SuperWare (R8 000)
The new market value of Intelligent is: 100 x 40 = R4 000
Your total investment in these two shares is now:
R8 000 + R4 000 = R12 000

New $w_{\text{SuperWare}}$: R8 000 / R12 000 = 0.667
New $w_{\text{Intelligent}}$: R4 000 / R12 000 = 0.333

$R_P = 0.667(0.15) + 0.333(0.20) = 0.16665 = 16.67\%$

$\sigma_P^2 = 0.667^2(0.08)^2 + 0.333^2(0.2)^2 + 2(0.667)(0.333)(0.29)(0.08)(0.20)$
$= 0.0009344$
$\sigma_P = (0.009344)^{½}$
$= 0.09666$
$= 9.67\%$

5. Which of the following portfolios would offer investors the best diversification benefits?

Portfolio	Portfolio constituents
A	Pick ‘n Pay Stores Ltd, The SPAR Group Ltd & Shoprite Holdings Ltd
B	Northam Platinum Ltd, Sephaku Holdings Ltd & DRDGold Ltd
C	JD Group Ltd, The Lewis Group Ltd & Italtile Ltd
D	Shoprite Holdings Ltd, Northam Platinum Ltd & JD Group Ltd

Solution

Portfolio D would be the best as it contains shares from three different industries. Although the returns of the four shares are likely to be positively correlated, the correlation coefficient is likely to be the lowest of the four options.

6. You won a portfolio equally invested in a risk-free asset and two shares. If one of the shares has a beta of 1.3 and the total portfolio is equally as risky as the market, what should the beta be on your portfolio?

The beta of a portfolio is the sum of the weight of each asset times the beta of each asset. If the portfolio is as risky as the market it must have the same beta as the market. Since the beta of the market is one, we know the beta of our portfolio is one. We also need to remember that the beta of the risk-free asset is zero. It has to be zero since the asset has no risk. Setting up the equation for the beta of our portfolio, we get:

$$\beta_p = \sum_{j=1}^{n} w_j \beta_j$$

$\beta_p = 1.0 = {}^1/_3(0) + {}^1/_3(1.3) + {}^1/_3(\beta x)$

Solving for the beta of Share X, we get:

$$1 = 0 + 0.43 + {}^{1}/_{3}(\beta x)$$
$$0.57 = {}^{1}/_{3}(\beta x)$$
$$\beta x = 0.57 \times 3$$
$$= 1.71$$

7. You are considering whether to invest in the following portfolio which consists of four shares. The invested amounts and beta coefficients of each of the shares are presented in the table below.

Share	Invested amount	Beta coefficient
African Coal Mining Ltd	R 400 000	0.77
Prime Gold Ltd	R 600 000	0.45
All Platinum Ltd	R1 000 000	1.53
AB Group Ltd	R2 000 000	1.24
Total	R4 000 000	

The rate of return on the FTSE/JSE All Share Index is 12% and the risk-free rate on government bonds 5.34%. If you forecast a 15% rate of return on the portfolio should you invest? Your forecast is based on your view of the economy.

Solution

Step 1: Calculate the portfolio beta

$$\beta_p = \sum_{j=1}^{n} w_j \beta_j$$

Share	Investment	Weight	Beta	$w_j \times \beta_j$
African Coal Mining Ltd	R 400 000	0.10	0.77	0.077
Prime Gold Ltd	R 600 000	0.15	0.45	0.068
All Platinum Ltd	R1 000 000	0.25	1.53	0.383
AB Group Ltd	R2 000 000	0.50	1.24	0.620
Total	R4 000 000	1.00		$\beta_p = 1.15$

Step 2: Use the SML equation to determine the portfolio's expected or required rate of return:

$$E(R_i) = R_f + \lfloor E(R_M) - R_f \rfloor x\ \beta_i$$
= 5.34 + (12 – 5.34)(1.15) = 13.00%

Since your forecasted rate of return (15%) > the SML's required rate of return (13.00%) you **should invest** in the portfolio.

PART ONE: FUNDAMENTALS OF FINANCIAL MANAGEMENT

CHAPTER 7: THE TIME VALUE OF MONEY

INTRODUCTION

This chapter covers the future value and present values of a lump sum value, missed stream amounts and annuities using annual and intra-year compounding. A number of variations are addressed namely calculating the deposits needed to accumulate a future sum, the amortisation of loans, the determination of interest or growth rates and the calculation of effective interest rates. The chapter concludes with how and where the time value of money is used within the field of financial management.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- the future value of a single amount
- the future value of an annuity
- the present value of a single amount
- the present value of a mixed stream of cash flows
- the present value of an annuity
- the present value of a perpetuity
- the present value of a growing perpetuity
- the deposits required to accumulate a future sum
- the instalments required to amortise a loan
- growth rates
- effective interest rates.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. An insurance company promises to pay Jane R1 million on her 60th birthday in return for a one-time payment of R125 000 today. (Jane just turned 30.) At what rate of interest would Jane be indifferent between accepting the company's offer and investing the premium on her own?

(a) 3.25%
(b) 5.16%
(c) 6.12%
(d) 7.18%
(e) The correct answer is not listed.

2. You are considering investing R400 in a 12-year annuity. The rate of return you require is 9%. What annual cash flow from the annuity will provide the required return?

(a) R10.77
(b) R42.96
(c) R55.86
(d) R78.31
(e) The correct answer is not listed.

3. You borrowed R1 200 at 8% compounded annually. Your payments are R168 at the end of each year. How many years will you make payments on the loan?

(a) 9 years
(b) 10 years
(c) 11 years
(d) 12.5 years
(e) The correct answer is not listed.

4. You have R100 000 to invest. Your bank offers one-year certificates of deposit with a stated rate of 3.50% compounded quarterly. What rate compounded semi-annually would provide you with the same amount of money at the end of one year?

(a) 3.485%
(b) 3.500%
(c) 3.505%
(d) 3.515%
(e) The correct answer is not listed.

5. Five years from now you will begin to receive cash flows of R125 per year. These cash flows will continue forever. If the discount rate is 10%, what is the present value of these cash flows?

(a) R776.15
(b) R894.22
(c) R934.07
(d) R1 136.36
(e) The correct answer is not listed.

Answers to short questions

1. d
2 c
3 a
4 d
5 a

EXAMPLES OF OPEN QUESTIONS

1. You received a R1 000 savings account earning 6% per annum on your 1st birthday. How much will you have in the account on your 21st birthday if you haven't withdrawn any money before then?

Solution

Calculator inputs: $n = 20$; $i = 6$; $PV = 1\ 000$; compute $FV = R3\ 207.14$

Discount factor:
$$\begin{aligned} FV &= PV \times FV_{r,t} \\ &= 1\ 000 \text{ x } (3.2071) \\ &= R3\ 207.10 \end{aligned}$$

Equation:
$$\begin{aligned} FV &= PV \times (1+r)^n \\ &= 1\ 000 \times (1.06)^{20} \\ &= R3\ 207.14 \end{aligned}$$

2. What is the future value of R15 000 received today if it is invested at 7.5% compounded annually for five years?

Solution

Calculator inputs: $n = 5$; $i = 7.5$; $PV = 15\ 000$; compute $FV = R21\ 534.44$

Discount factor:
$$\begin{aligned} FV &= PV \times FV_{r,t} \\ &= 15\ 000(1.43595) \\ &= R21\ 539.25 \end{aligned}$$

Equation:
$$\begin{aligned} FV &= PV \times (1+r)^n \\ &= 15\ 000 \times (1.075)^5 \\ &= R21\ 534.44 \end{aligned}$$

3. The Simple Bank pays 5% annum simple interest on its savings account balances, while the Complex Bank pays 5% interest per annum compounded annually. If you made a R15 000 deposit in each bank account, how much additional money would you earn from the 1st Complexity Bank account at the end of 10 years?

Solution

$$\begin{aligned} \text{Simple Bank interest} &= (0.05)(15\ 000)(10) = R7\ 500 \\ \text{Complex Bank interest} &= R15\ 000(1.05)^{10} - R15\ 000 \\ &= R9\ 433.42 \end{aligned}$$

Will receive an additional amount of R1 933.42

4. You will receive a R250 000 inheritance in 25 years. You can invest that money today at 8% compounded annually. What is the present value of your inheritance?

Solution

Calculator inputs: $n = 25$; $FV = 250\ 000$; $i = 8$; compute $PV = R36\ 504.48$

5. How much would you have to invest today at 9% compounded annually to have R35 000 available for an overseas trip five years from now?

Solution

Calculator inputs: n = 5; FV = -35 000; i = 9; compute PV = R22 747.60

6. The Cash Company Ltd has identified an investment project with the following cash flows. If the discount rate is 10%, what is the present values of these cash flows? What is the present value at 18%? At 24%?

YEAR	CASH FLOW
1	R1 500
2	R600
3	R855
4	R1 480

Solution

PV@10%
= R1 500 / 1.10 + R600 / 1.10^2 + R855 / 1.10^3 + R1 480 / 1.10^4 = R3 512.74
PV@18%
= R1 500 / 1.18 + R600 / 1.18^2 + R855 / 1.18^3 + R1 480 / 1.18^4 = R2 985.84
PV@24%
= R1 500 / 1.24 + R600 / 1.24^2 + R855 / 1.24^3 + R1 480 / 1.24^4 = R2 674.33

Alternatively use your financial calculator. As we have non-constant cash flows, we need to use the "cash flow register" or CF_i button (for SHARP calculators)

After you have cleared the memory, type in the following
(Steps for the SHARP EL738 Business/Financial calculator):

0 ENT
(NB: Your calculator ALWAYS sees the first entry into the cash flow register as the cash flow in Year 0. As there is no cash outflow in this exercise, you need to type 0)
1 500 ENT
600 ENT
855 ENT
1 480 ENT
2nd F CF_i
10 ENT for entering the required rate of return
ARROW DOWN COMP NET_PV = R3 512.74

NB: Since we use the cash flow register, you need to calculate Net Present Value (NPV) and not Present Value (PV). You will also notice that we have not entered the number of periods we are working with. The calculator identified the period. If you are using an older or other type of calculator, you may need to enter the period, which is 4 for this specific question.

In this question we don't want to clear the memory as we don't want to retype everything, just the new discount rate i.e. press ARROW UP 18 ENT, ARROW DOWN COMP NET_PV = R2 985.84. Once again without clearing the memory type ARROW UP 24 ENT, ARROW DOWN COMP NET_PV = R2 674.33.

7. Determine the present value of an investment offering the following cash inflows:

Year	Cash flows
1 – 5	R20 000 per annum
6 – 10	R35 000 per annum

The project costs R15 000 and the cash flows can be invested at an interest rate of 14%.

Solution

REMEMBER TO CLEAR YOUR CALCULATOR'S CASH FLOWS (CF_i)
Calculator inputs:
-15 000 ENT (NB: Type this as a negative as it represents a cash outflow).
20 000 ENT ENT ENT ENT ENT
(There are five cash inflows to the value of R20 000)
35 000 ENT ENT ENT ENT ENT
(There are five cash inflows to the value of R35 000)
n = 10 (enter only if calculator is not identifying the number of periods)
2nd F CFi
14 ENT
ARROW DOWN COMP NET_PV = R116 067.83

8. Assume that you have been employed for a couple of years and own a flat in the Summerstrand area. Your cousin, a BCom Honours student rented the flat in 2022. You collected the rent of R3 500 at the end of each month for 10 months. You deposited this money in an account paying 0.5% interest per month. As your cousin moved out, you rented the flat to an international student in 2024 and require that rent be paid at the beginning of each month for 10 months. How much money did you lose by letting the flat to your cousin, assuming that the interest rate remained unchanged?

Solution

- FV of funds collected from your cousin: **Make sure your calculator is on END mode! The default setting of your calculator is END mode.**
  n = 10 months; i = 0.5% per month; PMT = R3 500 per month;
  Compute FV = **R35 798.09**
- FV of funds collected from the international student: **NB: Press BGN!!!!**
  n = 10 months; i = 0.5% per month; PMT = R3 500 per month;
  Compute FV = **R35 977.08 NB: Press BGN again to de-activate it.**
  Difference between FV of international student and cousin:
  35 977.08 – 35 798.09 = **R178.99**

9. An account was opened with an investment of R2 000 ten years ago. The ending balance in the account is R3 000. If interest was compounded annually, what rate was earned on the account?

Solution

Calculator inputs: n = 10; PV = -2 000; FV = 3 000; compute i = 4.14%

10. You've purchased shares in the RAPS Ltd in February 2016. The RAPS Group Ltd acts as a wholesale and distributor of goods and services to RAPS supermarkets,

building materials outlets and liquor stores. You have purchased 5 000 shares at a price of R64.70. Considering the growth of the share price of the last few years, you expect the price to be R120.00 in February 2026. What is the growth rate in your wealth, should your expectation about the share price be correct?

Solution

Shareholder wealth = no of issued shares x share price at a particular point in time

Date	No of shares	Share price	Wealth
February 2016	5 000	R64.70	R323 500
February 2026	5 000	R120.00	R600 000

Calculator inputs: n = 10; PV = -323 500; FV = 600 000; compute i = 6.37%
Since the number of shares stays the same, you could also only look at the change in share price…

Calculator inputs: n = 10; PV = -64.70; FV = 120.00; compute i = 6.37%

11. The First Bank charges 14.8% per annum, compounded quarterly, on its business loans. The Second Bank charges 15.0% per annum compounded semi-annually. As a potential borrower, which bank would you go to for a new loan?

Solution

Last National Bank: EAR = $[1 + (0.158 / 4)]^4 - 1$ = 15.64%
Last United Bank: EAR = $[1 + (0.15 / 2)]^2 - 1$ = 15.56%
The loan from Last United Bank is cheaper (not much though …).

12. You would like to purchase a house. The prime interest rate is 9%. Your bank is willing to give you a loan for 25 years at a fixed interest rate of prime + 1%. Calculate the monthly repayments on the Kamma Height residence which you have identified from cyberprop.com. Ignore transfer duties, taxes, estate agent fees etc.

Suburb	**Kamma Heights**	**Mill Park**
Price	ZAR1.6million	ZAR2.4million
Notes	Modern living	Charm, Convenience & Prestige...!! 303m² of double storey splendour!
Bedrooms	3	3
Bathrooms	2	2
Main en Suite	-	1
Garages	2	2
Garden	Beautiful Garden	Yes
Floor Type	Carpets	Heated carpets
Electronic gates	Yes	Yes
Intercom	Yes	Yes
Pool	Yes	Yes
Security	Yes	Yes

Solution

Calculator inputs for Kamma Heights residence:
PV = -1 600 000; n = 25 x 12 = 300; i = 10 / 12 = 0.8333;
compute PMT = R14 539.21 per month
(Hopefully you have a good salary, have inherited lots of money or married rich…)

13. Construct an amortisation table in Microsoft Excel for the Mill Park residence costing R2.4million. The loan term is 25 years and the bank will charge a fixed interest rate of 10% (e.g. prime + 1%) compounded annually.

Solution

Loan amount	2 400 000
Interest rate	10
Loan term (years)	25
Annual loan payment[(a)]	264 403.37

Calculator inputs: n = 25; i = 10; PV = -2 400 000;
compute PMT = R264 403.37

Year	Beginning balance	Total payment	Interest (10%)	Principle paid	Ending balance
1	R 2 400 000.00	R 264 403.37	R 240 000.00	R 24 403.37	R 2 375 596.63
2	R 2 375 596.63	R 264 403.37	R 237 559.66	R 26 843.71	R 2 348 752.92
3	R 2 348 752.92	R 264 403.37	R 234 875.29	R 29 528.08	R 2 319 224.83
4	R 2 319 224.83	R 264 403.37	R 231 922.48	R 32 480.89	R 2 286 743.94
5	R 2 286 743.94	R 264 403.37	R 228 674.39	R 35 728.98	R 2 251 014.97
6	R 2 251 014.97	R 264 403.37	R 225 101.50	R 39 301.88	R 2 211 713.09
7	R 2 211 713.09	R 264 403.37	R 221 171.31	R 43 232.06	R 2 168 481.02
8	R 2 168 481.02	R 264 403.37	R 216 848.10	R 47 555.27	R 2 120 925.75
9	R 2 120 925.75	R 264 403.37	R 212 092.58	R 52 310.80	R 2 068 614.96
10	R 2 068 614.96	R 264 403.37	R 206 861.50	R 57 541.88	R 2 011 073.08
11	R 2 011 073.08	R 264 403.37	R 201 107.31	R 63 296.07	R 1 947 777.01
12	R 1 947 777.01	R 264 403.37	R 194 777.70	R 69 625.67	R 1 878 151.34
13	R 1 878 151.34	R 264 403.37	R 187 815.13	R 76 588.24	R 1 801 563.10
14	R 1 801 563.10	R 264 403.37	R 180 156.31	R 84 247.06	R 1 717 316.04
15	R 1 717 316.04	R 264 403.37	R 171 731.60	R 92 671.77	R 1 624 644.27
16	R 1 624 644.27	R 264 403.37	R 162 464.43	R 101 938.95	R 1 522 705.32
17	R 1 522 705.32	R 264 403.37	R 152 270.53	R 112 132.84	R 1 410 572.48
18	R 1 410 572.48	R 264 403.37	R 141 057.25	R 123 346.12	R 1 287 226.36
19	R 1 287 226.36	R 264 403.37	R 128 722.64	R 135 680.74	R 1 151 545.62
20	R 1 151 545.62	R 264 403.37	R 115 154.56	R 149 248.81	R 1 002 296.81
21	R 1 002 296.81	R 264 403.37	R 100 229.68	R 164 173.69	R 838 123.12
22	R 838 123.12	R 264 403.37	R 83 812.31	R 180 591.06	R 657 532.06
23	R 657 532.06	R 264 403.37	R 65 753.21	R 198 650.17	R 458 881.89
24	R 458 881.89	R 264 403.37	R 45 888.19	R 218 515.18	R 240 366.70
25	R 240 366.70	R 264 403.37	R 24 036.67	R 240 366.70	R 0.00

PART ONE: FUNDAMENTALS OF FINANCIAL MANAGEMENT

CHAPTER 8: VALUATION OF SHARES AND DEBENTURES

INTRODUCTION

This chapter discusses the valuation of capital sources, both shares and debentures. It defines value, highlights the key input of valuation, namely timing and the discount rate. The chapter mentions the basic valuation model and then details the valuation of debentures and the various share valuation models.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- distinguish between different types of value
- determine value based on the discounted cash flow (DCF) technique
- determine the value of debentures
- determine the yield to maturity (YTM)
- determine the value of ordinary shares.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. Which one of the following statements is *correct*?

(a) The book value reflects the current value of an asset.
(b) The market value reflects the current value of an asset.
(c) The future value reflects the current value of an asset.

2. Jewellers Pandora acquired an asset that is expected to generate cash flows of R2 500, R0, R3 000, R9 800 at the end of Years 1, 2, 3 and 4 respectively. The firm's required rate of return is 16%. The value of the asset is

(a) R15 300.
(b) R5 500.
(c) R4 300.
(d) R9 749.64.
(e) R9 489.60.

3. The debentures of Super Pots Ltd will be issued at a par value of R1 000 each. Interest is paid annually, and the required rate of return is equal to the debenture's coupon interest of 12%. The initial maturity is 10 years. The debenture will be a __________ debenture.

(a) premium
(b) discount
(c) par value

4. Use the information given in the previous question. The value of a Super Pots Ltd debenture is

(a) R1 000.
(b) R1 200.
(c) R10 000.
(d) R12 000.
(e) R120.

5. The Shoe World Company has a required rate of return of 14%. It is considering investing in debentures that will be issued at a par value of R1 000, with a coupon interest of 12% (paid semi-annually) and an initial maturity of 20 years. The value of the debenture is

(a) R880.68.
(b) R1 140.
(c) R1 200.
(d) R866.68.
(e) R1 000.

Answers to short questions

1. b
2. e
3. c
4. a
5. d

EXAMPLES OF OPEN QUESTIONS

1. Would it be an appropriate time to invest in shares once the monetary policy committee announces a decline in their repo rate (the rate they charge banks for borrowing from them)?

2. The Super Slush Ice Cream company just paid a dividend of R4.00 per share. The company will increase its dividend by 10% next year. Thereafter, this dividend growth rate will reduce to the industry average of 5% per year, and this constant growth rate will be kept forever. If the required rate of return on Super Slush Ice Cream is 11%, what will one of its shares sell for today? Work with TWO decimal places <u>throughout</u> this question.

Solution

Step 1: Dividends DURING first growth period

D_0 = 4
D_1 = D_0 (1 + g)
= 4 (1.1) = 4.40

Step 2: PV of dividends DURING first growth period

PV of D_1: FV -4.40; n = 1; i = 11; compute PV = 3.96
Sum = 3.96

Step 3: Value of the constant dividends AFTER first growth period (thus based on second period)

P_1 = D_2 / (k_s – g)
= 4.62 / (0.11 – 0.05)
= 77.00

D_2 = D_1 (1 + g) REMEMBER: g = 5% now!!!!!
= 4.4 (1.05)
= 4.62

Step 4: PV of all the constant dividends AFTER first growth period (thus based on second period)

PV of P_1: FV -77.00; n = 1; i = 11; compute PV = 69.37

Step 5: Add PV's of step 2 and step 4

3.96 + 69.37 = R73.33

Thus, current price or value of the share (P_0) = R73.33

PART TWO: SHORT-TERM FINANCIAL MANAGEMENT: THE MANAGEMENT OF WORKING CAPITAL

CHAPTER 9: NET WORKING CAPITAL AND CASH FLOW MANAGEMENT

INTRODUCTION

This chapter covers topics relating to net working capital and the cash flow management of firms. The chapter discuss financing policies and sources of short-terms financing. In particular, strategies and techniques for cash management and using a cash budget for cash planning are highlighted.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- decide on a working capital financing policy
- identify sources of short-term financing
- distinguish between profitability and cash flow
- efficiently manage cash
- estimate optimal cash balances
- compile a cash budget
- prevent losses of cash by means of internal control.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. Which one of the following statements is *incorrect*?

(a) Short-term debt is more expensive than long-term debt.
(b) The approach that aims to match a firm’s asset and liability maturities is the moderate working capital policy.
(c) The use of short-term debt rather than long-term debt may increase income margins.
(d) The aggressive approach is the riskier approach in terms of working capital financing.
(e) The interest rates on short-term financing may fluctuate rapidly and increase to very high levels.

2. How many of the following statements are correct?

   (i) When a firm stretches its account payable, it pays bills as late as possible.
   (ii) With better purchasing planning a firm can reduce the length of its purchasing cycle and thus increase its inventory turnover or average age of inventory.
   (iii) A firm should aim to increase its average collection period since this will decrease its cash conversion cycle.
   (iv) A firm should strive for a negative cash conversion cycle.

(a) none
(b) one
(c) two
(d) three
(e) four

3. Global Sports Ltd borrow R50 000 at a 12% interest rate for a year, and the interest is payable in advance. The effective interest rate that the firm is paying is

(a) R5 280.
(b) R6 000.
(c) 11.75%.
(d) 12.00%.
(e) 13.64%.

4. Johnson Wear Ltd is offered trade credit terms of 3/15, net 45. This means

(a) the firm receives a 15% discount if the invoice is paid within 3 days of purchase, otherwise if the discount is not taken the firm should pay in full within 45 days of purchase.
(b) the firm receives a 3% discount if the invoice is paid within 15 days of purchase, otherwise if the discount is not taken the firm should pay in full within 45 days of purchase.
(c) the firm receives a 3% discount if the invoice is paid within 45 days of purchase, otherwise if the discount is not taken the firm should pay in full within 15 days of purchase.

5. Calculate the cost to Johnson Wear Ltd if it foregoes the following cash discount: 3/15, net 45, assuming a 360-day year.

(a) 3.11%.
(b) 15.15%.
(c) 15.50%.
(d) 37.11%.
(e) 45.00%.

Answers to short questions

1. a
2. c
3. e
4. b
5. d

EXAMPLES OF OPEN QUESTIONS

1. Stationary Galore Ltd sells all its merchandise on credit. The firm has established that it takes and average of 70 days between purchasing a product, stocking it and then selling it. However, when customers buy the products they receive credit, and on average debtors' accounts are collected after 62 days. The firm also purchases its inventory from suppliers on credit. On average, the firm's payment period to its creditors is 40 days. Calculate and interpret the firm's cash conversion cycle.

Solution

Cash conversion cycle (CCC) = AAI + ACP – APP
= 70 + 62 – 40
= 92 days

The firm is receiving cash on average 92 days after it pays its bills. It thus needs to secure short-term financing for just over 3 months.

2. The company Wonder Car Parts had sales of R600 000 in April and R650 000 in May. Forecast sales for June, July and August are R700 000, R780 000 and R810 000, respectively. The firm has a cash balance of R55 000 on 1 June and wishes to maintain a minimum cash balance of R55 000. Given the following data, prepare and interpret a cash budget for the months of June, July and August:

- 20% of the firm's sales are for cash, 70% is collected in the next month and the remaining percentage is collected in the second month following the sale.
- The firm receives other income of R5 000 per month.
- The firm's expected purchases, all made for cash, are R500 000, R520 000 and R480 000 for June, July and August, respectively.
- Lease payments are expected to amount to R25 000 per month.
- Wages and salaries are 10% of the previous month's sales.
- Cash dividends of R45 000 will be paid in July.
- Payment of principal and interest of R22 000 is due in July.
- A cash purchase of equipment costing R630 000 is scheduled for November.
- Taxes of R85 000 are due in June.

Solution

WONDER CARS PARTS
CASH BUDGET FOR THE PERIOD JUNE TO AUGUST 2025

	R	R	R
	June	**July**	**August**
Cash receipts	655 000	711 000	778 000
Other income	5 000	5 000	5 000
Total receipts	**660 000**	**716 000**	**783 000**
Payment for purchases	500 000	520 000	480 000
Lease	25 000	25 000	25 000
Salaries and wages	65 000	70 000	78 000
Dividends	-	45 000	-
Loan repayments and interest	-	22 000	-
Tax	85 000	-	-
Total disbursements	**(675 000)**	**(682 000)**	**(583 000)**
NET CASH FLOW	**(15 000)**	**34 000**	**200 000**
Cash at the beginning of the period or minimum	55 000	55 000	74 000
Cash at the end of the period before borrowing	**40 000**	**89 000**	**274 000**
Financing needed	15 000	-	-
Cash at the end of the period	55 000	89 000	274 000
Cumulative borrowing	15 000	(15 000)	(15 000)
EXCESS CASH	**-**	**74 000**	**259 000**

The firm has a cash shortfall during the month of June and needs short-term borrowing for this period. During the months of July and August there is a steady increase in the cash flow of the firm and there are excess cash balances during both these months.

PART TWO: SHORT-TERM FINANCIAL MANAGEMENT: THE MANAGEMENT OF WORKING CAPITAL

CHAPTER 10: THE MANAGEMENT OF ACCOUNTS RECEIVABLE

INTRODUCTION

This chapter covers the procedure of allowing and managing credit. Therefore, this chapter focuses on the establishment of the credit policy, the process of following up on outstanding accounts and monitoring accounts receivables.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- establish a credit policy for a form
- devaluate the influence of changes in credit terms on profitability
- monitor and follow up accounts receivable.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. All of the following are aspects that need to be taken into account when establishing a credit policy, **EXCEPT**:

(a) credit selection.
(b) credit standards.
(c) credit limits.
(d) credit terms.
(e) credit payments.

2. _________are professional businesses that provide services for a predetermined fee to firms seeking information on credit ratings of potential customers.

(a) Credit insurance
(b) Bank references
(c) Credit agencies
(d) Trade sources
(e) Trade references

3. Which ONE of the following is NOT one of the five Cs when conducting a credit analysis?

(a) Capital
(b) Conditions
(c) Character
(d) Collateral
(e) Capability

4. A customer has been given the credit terms of receiving a 3% discount if the invoice is paid within 15 days from date of invoice, with the invoice to be paid in full within 30 days. In addition, interest of 5% will be charged on overdue accounts. The credit terms will thus be:

(a) 5/15 net 30 days
(b) 3/15 net 30 days
(c) 2/15 net 30 days
(d) 3/15 net 15 days
(e) 3/15 net 45 days

5. A company has bad debts of R250 000 at the end of the month. Sales (credit and cash) of R1 100 000 was reported for the month with 70% of the sales being on credit. The credit purchases for the month amounted to R600 000. Calculate the bad debt ratio applicable for this month.

(a) 22.73%
(b) 32.47%
(c) 41.67%.
(d) 54.55%.
(e) 75.76%

Answers to short questions

1. e
2 c
3 e
4 b
5 b

EXAMPLES OF OPEN QUESTIONS

1. Extreme bicycles, a manufacturer of high quality, light weight aluminium bicycles who operates on a cash-only basis, received a request from several clients to extend credit to "net one month (30 days)." The selling price per unit is R2 500, the variable cost per unit is R1 400, the current quantity is 3 000 per month, while the quantity sold under the new policy will increase to 3 400 per month and the expected average collection period will be 36 days. Bad debts are expected to amount to 0.5% of credit sales. The required rate of return is 20% per month.

Determine effect on profit:
Increase in profits = (3 400 – 3 000) x (R2 500 – R1 400)
= R440 000

Cost of average investment in accounts receivable

Variable cost of existing policy	= (R1 400 x 3 000)	
	= R4 200 000	
Variable cost of new policy	= (R1 400 x 3 400)	
	= R4 760 000	
Turnover of accounts receivable under existing policy		= 0 (cash only)
Turnover of accounts receivable under new policy		= 360 / 36
		= 10 times

Marginal investment in accounts receivable

Existing policy	= 0
New policy	= R4 760 000 / 10
	= R476 000
Marginal investment	= (R476 000)
Cost of marginal investment	= 476 000 x 0.2
	= R95 200

Cost of marginal bad debts

Bad debts under existing policy	= 0
Bad debts under new policy	= 0.005 x 2 500 x 3 400
	= R42 500

Bad debts will increase from zero to R42 500

Cost of settlement: no discount given under existing or new policy.

Final decision

Increase in profits	R440 000
Cost of marginal investment	R 95 200
Cost of bad debts	R 42 500
Net increase in profit under new policy	R302 300

Accept the new policy as the net profit will increase under the new policy.

PART TWO: SHORT-TERM FINANCIAL MANAGEMENT: THE MANAGEMENT OF WORKING CAPITAL

CHAPTER 11: THE MANAGEMENT OF INVENTORY

INTRODUCTION

This chapter focuses on the importance of having adequate inventory, the various classes of inventory and the classification of inventory using the ABC method. Four techniques to evaluate inventory as well as the various cost of inventory that should be taken into account, are discussed. Quantity discounts and inflation are also important to take into account. In the management of inventory, various inventory control systems can be implemented to monitor inventory balances and losses.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- classify inventory
- do a valuation of inventory
- calculate the economic order quantity
- determine the carrying, ordering and shortage costs of inventory
- determine the reorder point
- perform an ABC classification of inventory
- manage safety stock and inventory balances
- monitor inventory turnover rates
- prevent loss of inventory.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. All of the following are reasons for carrying raw material (inventory), **EXCEPT**:

(a) to make the scheduling of production easier.
(b) to hedge against price changes.
(c) to hedge against quality issues.
(d) to make use of quantity discounts.
(e) to hedge against supply shortages.

2. The value of the inventory on hand is R15 000 while the total carrying cost amounts to R2 550. The price per unit is R50 and the average inventory on hand is 200. There is no safety stock. What is the total annual carrying cost?

(a) R1 700
(b) R5 625
(c) R6 667
(d) R8 367
(e) R10 000

3. The firm requires 800 bottles per month and carries an average inventory of 250 bottles. The cost per order is R25. What is the total annual ordering cost?

(a) R40
(b) R80
(c) R330
(d) R417
(e) R480

4. Calculate the reorder point using the following information:

Price per unit	R50
Cost per order	R15
Lead time in days	5
Daily sales (demand)	60

(a) 175 units
(b) 180 units
(c) 300 units
(d) 420 units
(e) 1 500 units

5. Safety stock levels vary from firm to firm and therefore all of the following factors should be considered, **EXCEPT**:

(a) inventory shortage cost.
(b) certainty of supply forecasts.
(c) carrying cost of additional inventory.
(d) probability of delays in delivery.
(e) certainty of demand forecasts.

Answers to short questions

1. c
2. a
3. e
4. c
5. b

EXAMPLES OF OPEN QUESTIONS

1. Suppose that a jewellery manufacturer, expects that annual consumption of a certain item to be 7 200 units and the average cost per order is R55. The purchase price is R20 per item and the average inventory holding costs is 30% per annum. Calculate the firm's economical order quantity (EOQ).

Solution

$$\text{Economic ordering quantity} = \text{EOQ} = \sqrt{\frac{2 \times O \times T}{C \times PP}}$$

$$= \sqrt{(2 \times 55 \times 7\,200) / (0.30 \times R20)}$$

$$= \sqrt{792\,000 / 6}$$

= 363 units

The most economical quantity to order every time the business places an order of this specific item is 363 units.

2. The following information is available regarding the order lead time multipliers and stock-out consequence multipliers of inventory:

Lead time class	Multiplier
0-2 day	0
3-15 days	2
16-30 days	4
Longer than 30 days	0

The stock-out consequence multipliers are the following:

Unimportant	1
Average	15
Critical	30

Calculate the management importance of inventory with a quantity of 50 000 bottles at a cost of R25 per bottle. The lead time of this specific item is 18 days with this item being classified as critical.

Solution

Value = 50 000 x 25 x (4 + 30)
= R42 500 000

PART THREE: LONG-TERM FINANCIAL MANAGEMENT: INVESTMENTS

CHAPTER 12: CAPITAL BUDGETING AND CASH FLOW PRINCIPLES

INTRODUCTION

This chapter covers capital budgeting and cash flow principles in order to show students the process of capital budgeting or capital investment analysis. This process is about evaluating and selecting long-term investments consistent with the goal of increasing a firm's value. The chapter considers the motives and steps of the capital budgeting process as well as various project types and cash flow types. Thereafter, the chapter focus on the relevant cash flows to take into account when evaluating projects.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- distinguish between the motives for capital expenditure
- apply the capital budgeting process
- distinguish between independent and mutually exclusive projects
- calculate the initial investment, and operating and terminal cash flows of a proposed capital expenditure projects.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. A firm identified two options to increase its production capacity, namely, to appoint another production team of 10 employees, alternatively to equip its factory with two Artificial Intelligence (AI) machines. These options

(a) are independent.
(b) are mutually exclusive.
(c) will require no changes in net working capital.
(d) will only show financial implications one production starts.
(e) will only influence cash inflows.

2. Fishing Rods Ltd considers an investment that will result in an initial cash outflow of R50 000. Thereafter, the investment is expected to realise cash inflows of R20 000, R35 000, R35 000 and R15 000 over the next four years respectively. This investment will be regarded as a(n)

(a) annuity.
(b) unconventional investment.
(c) mixed stream investment.
(d) once-off investment.
(e) terminal.

3. **TRUE OR FALSE:** Installation costs are irrelevant and do not form part of a project's initial investment.

(a) True
(b) False

4. **TRUE OR FALSE:** Usually, when calculating a project's terminal cash flow, the change in net working capital reverses the net working capital investment that occurred at initial investment, and it will be shown as a cash inflow when determining the terminal cash flow.

(a) True
(b) False

5. A firm is evaluating an investment. In calculating the expected cash flow for each year (operating cash flows), one has to add back depreciation because it is

(a) a cash inflow.
(b) part of the initial investment.
(c) a liability.
(d) proceeds from the sale of an asset.
(e) not a real cash outflow.

Answers to short questions

1. b
2. c
3. b
4. a
5. e

EXAMPLES OF OPEN QUESTIONS

1. A firm is considering replacing equipment and investing in technologically advanced equipment. The new equipment will cost R5 500 000 and installation costs will amount to R500 000. The firm uses straight-line depreciation, and the new equipment may be depreciated over a period of six years. The old equipment has a book value of R0 but can be sold for R60 000. The firm is subject to a 27% tax rate. The new equipment will be used for six years and is expected to generate the following increases in net operating profits after tax (NOPAT):

Year	Expected increase in NOPAT
1	1 250 000
2	1 250 000
3	1 250 000
4	1 000 000
5	1 000 000
6	900 000

At the end of Year 6, the firm expects to sell the equipment (with a book value of 0) for R30 000. Net capital worth R1 300 500 incurred when project is excepted, is expected to be recovered at the end of the investment project.

Use the template below to calculate the total cash flows for each year.

	Year 0	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**	**Year 6**
Component 1: Initial investment (II)							
Component 2: OCFs							
Component 3: Terminal CF							
Total cash flows							

Solution

Workings (NOTE: all cash outflows shown as negative values and all cash inflows as positive values)

Component 1: II

– Cost of project	5 500 000	
− Installation cost	500 000	
+ Proceeds from sale of old equipment	60 000	
− Tax liability	16 200	
+ Change in NWC	1 300 500	(in this case change is an outflow)
	–7 256 700	

Component 2: OCF
Each year's expected increase in NOPAT (as given in table on p 55), plus (add back) the depreciation per annum (as it is a non-cash item)
Depreciation per annum: (5 500 000 + 500 000) / 6 = 1 000 000

Component 3: Terminal cash flow

+ Proceeds from the sale of equipment	30 000	
– Tax liability on the sale of equipment	8 100	
+ Change in NWC from Y_0	1 300 500	(reversal of NWC in component 1)
	1 322 400	

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Component 1: Initial investment (II)	-7 256 700						
Component 2: OCFs		2 250 000	2 250 000	2 250 000	2 000 000	2 000 000	1 900 000
Component 3: Terminal CF							1 322 400
Total cash flows	-7 256 700	2 250 000	2 250 000	2 250 000	2 000 000	2 000 000	3 222 400

2. A project will produce operating cash flows of R45 000 a year for three years. At the start of the project, the current assets will increase by R15 000 and the current liabilities will decrease by R10 000. At the end of the project, net working capital will return to its normal level. The project requires the purchase of equipment at an initial cost of R120 000. The equipment will be depreciated straight-line to a zero book value over the life of the project. The equipment will be salvaged at the end of the project creating a R25 000 after-tax cash flow.

Use the template below to calculate the total cash flows for each year.

	Year 0	Year 1	Year 2	Year 3
Component 1: Initial investment (II)				
Component 2: OCFs				
Component 3: Terminal CF				
Total cash flows				

Solution

Workings

Component 1: II
– Cost of project 120 000
+ Change in NWC from Y_0 25 000* (in this case the change in the NWC is an outflow)
–145 000

*CA increase is a cash outflow and CL decrease is a cash outflow, thus
-15 000 – 10 000 = -25 000

Component 2: OCF
The OCF for the three years were given as R45 000 pa. No calculation needed to determine OCF.

Component 3: Terminal cash flow
+ After-tax value of asset sold 25 000
+ Change in NWC from Y_0 25 000
50 000

	Year 0	**Year 1**	**Year 2**	**Year 3**
Component 1: Initial investment (II)	-145 000			
Component 2: OCFs		45 000	45 000	45 000
Component 3: Terminal CF				50 000
Total cash flows	-145 000	45 000	45 000	95 000

PART THREE: LONG-TERM FINANCIAL MANAGEMENT: INVESTMENTS

CHAPTER 13: CAPITAL BUDGETING TECHNIQUES

INTRODUCTION

This chapter covers various techniques used to make capital budgeting or investments in long-term assets decisions. The approaches to decision-making are discussed, thereafter both the non-discounted cash flow methods and the discounted cash flow methods are shown. The chapter concludes with a discussion and NPV profiles and how to treat conflicting rankings.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- apply an appropriate approach to the capital budget decisions
- calculate the average rate of return and payback period of capital expenditure proposals
- calculate and interpret the net present value (NPV), profitability index (PI), and internal rate of return (IRR)
- evaluate projects with conflicting NPV rankings.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. A situation in which accepting one investment prevents the acceptance of another investment is called the

(a) net present value profile.
(b) operational ambiguity decision.
(c) multiple choices of operations decision.
(d) issues of scale problem.
(e) mutually exclusive investment decision.

2. Which one of the following is the best example of two mutually exclusive projects?

(a) Planning to build a warehouse and a retail outlet side by side.
(b) Buying sufficient equipment to manufacture both desks and chairs simultaneously.
(c) Using an empty warehouse for storage or renting it entirely out to another firm.
(d) Using the company sales force to promote sales of both shoes and socks.
(e) Buying both inventory and fixed assets using funds from the same bond issue.

3. The present value of the cash flow of an investment is expected to total R2 500 000. The initial investment is R1 300 000. The profitability index (PI) is

(a) 1.92.
(b) 0.52.
(c) 19.20%.
(d) R1 200 000.
(e) The Profitability Index is impossible to determine.

4. You have been appointed as the financial manager of Boots Forever (Pty) Ltd. You need to evaluate an investment project with the following cash flows:

Year	**Project A (CFs in Rands)**
0	-250 000
1	25 500
2	35 000
3	16 500
4	350 000

The applicable discount rate for this project is 15%. The firm also wishes to repay projects' initial costs within 4 years.

The Payback Period of Project A is

(a) 4 years.
(b) 3.95 years.
(c) 3.49 years.
(d) 3 years.
(e) The Payback Period is impossible to determine.

5. You have been appointed as the financial manager of Boots Forever (Pty) Ltd. You need to evaluate an investment project with the following cash flows:

Year	Project A (CFs in Rands)
0	-250 000
1	25 500
2	35 000
3	16 500
4	350 000

The applicable discount rate for this project is 15%. The firm also wishes to repay projects' initial costs within 4 years.

The Discounted Payback Period of Project A is

(a) 4 years.
(b) 3.95 years.
(c) 3.49 years.
(d) 3 years.
(e) The Discounted Payback Period is impossible to determine.

Answers to short questions

1. e
2. c
3. a
4. c
5. b

EXAMPLES OF OPEN QUESTIONS

1. You have been appointed as the financial manager of Marketing Blitz (Pty) Ltd. You need to evaluate two projects with the following cash flows:

Years	Project A (CFs in Rands)	Project A (CFs in Rands)
0	-250 000	-250 000
1	30 500	28 000
2	45 000	42 500
3	45 000	40 000
4	470 000	365 500

The applicable discount rate for both these two projects is 15% and the firm has a cut-off period of four years.

(a) Calculate the Net Present Value (NPV) of Project A. Will you accept or reject Project A based on its NPV? Motivate your answer.
(b) Calculate the Internal Rate of Return (IRR) of Project A. Will you accept or reject Project A based on its IRR? Motivate your answer.
(c) Calculate the Net Present Value (NPV) of Project B. Will you accept or reject Project A based on its NPV? Motivate your answer.
(d) Calculate the Internal Rate of Return (IRR) of Project B. Will you accept or reject Project A based on its IRR? Motivate your answer.

(e) Assume these two projects are independent. What will be your final decision?
(f) Assume these two projects are mutually exclusive. What will be your final decision?

Solution

(a)
CF0 = -250 000
CF1 = 30 500
CF2 = 45 000
CF3 = 45 000
CF4 = 470 000
i = 15
NPV = R108 860.46
Project A is acceptable since its NPV exceeds 0.

(b)
CF0 = -250 000
CF1 = 30 500
CF2 = 45 000
CF3 = 45 000
CF4 = 470 000
IRR = 27.67%
Project A is acceptable since its IRR (27.67%) exceeds the cost or capital or required rate of return (15%).

(c)
CF0 = -250 000
CF1 = 28 000
CF2 = 42 500
CF3 = 40 000
CF4 = 365 500
i = 15
NPV = R41 760.39
Project B is acceptable since its NPV exceeds 0.

(d)
CF0 = -250 000
CF1 = 28 000
CF2 = 42 500
CF3 = 40 000
CF4 = 365 500
IRR = 20.32%
Project B is acceptable since its IRR (20.32%) exceeds the cost or capital or required rate of return (15%).

(e)
Mutually exclusive means that more than one project can be selected, if acceptable. Thus, since both Projects A and B are acceptable on both criteria (NPV and IRR), the decision will be to accept both (thus to invest R500 000).

(f)
Independent projects mean that only one project – the best project that will add the most value to the firm – can be selected. Therefore, since Project A has both a higher NPV and higher IRR than Project B, the decision will be to accept only Project A.

2. Tara Limited is evaluating two mutually exclusive capital budgeting projects that have the following cash flows:

Years	Project Q (CFs in Rands)	Project R (CFs in Rands)
1	0	3 500
2	5 000	1 100

The cost of each of the two projects is R4 000.

Calculate and interpret the cross-over rate for the two projects.

Solution

Years	Project Q (CFs in Rands)	Project R (CFs in Rands)	CF difference
0	-4 000	-4 000	0
1	0	3 500	-3 500
2	5 000	1 100	3 900

CF0 = 0
CF1 =-R3 500
CF2 =R3 900
i or cross-over rate = 11.43%

At a discount rate (or required rate of return or cost of capital) of 11.43%, the firm will be indifferent about Projects Q and R. Their NPVs will be the same, and it won't matter which one of these projects is selected (if they meet the investment decision criteria based on the capital budgeting techniques), since they will add the same (amount of) value to the firm.

PART THREE: LONG-TERM FINANCIAL MANAGEMENT: INVESTMENTS

CHAPTER 14: RISK ADJUSTMENTS AND OTHER REFINEMENTS TO CAPITAL BUDGETING

INTRODUCTION

This chapter discusses the risk adjustments necessary for, and other situations that might be possibilities, to consider when evaluating capital projects (capital budgeting or long-term asset investments). In particular, the chapter focuses on decision tree, sensitivity and scenario analyses. Risk adjustment techniques are addressed. The capital budgeting refinements where projects with unequal lives are evaluated and when firms operate under conditions of capital rationing.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- evaluate which approach would be best to incorporate risk in the capital budgeting process
- incorporate risk in capital budgeting by means of certainty equivalents, decision tree analysis and risk-adjusted discount rates
- evaluate projects with unequal lives
- apply capital rationing to the capital budgeting process
- evaluate projects in view of budget constraints.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. _________ is used to evaluate the sensitivity of various capital investment proposals by adjusting the future cash flows depending how certain or uncertain these inflows are expected to be

(a) Sensitivity analysis
(b) Scenario analysis
(c) Certainty equivalents
(d) Decision tree
(e) Capital rationing

2. _________ is to evaluate capital investment proposals based on their IRR, but accepting only those with the greatest IRR values withing the budget constraint of the firm.

(a) Sensitivity analysis
(b) Scenario analysis
(c) Certainty equivalents
(d) Decision tree
(e) Capital rationing

3. The _________ is the evaluation of the sensitivity of various capital investment proposals by adjusting the discount rate by adding risk premiums.

(a) certainty equivalents
(b) NPV
(c) IRR
(d) RADR
(e) ANPV

4. The objective of _________ is to select the group of projects that is expected to yield the greatest total NPV and that does not require more fands than were budgeted for.

(a) sensitivity analysis
(b) scenario analysis
(c) certainty equivalents
(d) decision-tree
(e) capital rationing

5. Which one of the following is the correct equations to calculate the accounting break-even?

(a) (total fixed costs + depreciation) / (price per unit – variable cost per unit)
(b) (total fixed costs – depreciation) / (price per unit – variable cost per unit)
(c) (total costs + depreciation) / (price per unit – variable cost per unit)
(d) (total fixed costs + operating cash flow) / (price per unit – variable cost per unit)
(e) total fixed costs + depreciation / (price per unit – variable cost per unit)

Answers to short questions

1. c
2. e
3. d
4. e
5. a

EXAMPLES OF OPEN QUESTIONS

1 Dream Vacations Ltd is analysing a proposed project. The following information is available regarding the project:

Sales quantity	2 500 units
Selling price per unit	R16
Variable cost per unit	R 8
Fixed costs	R12 000

1.1 Complete the firm's budgeted/pro forma statement of financial performance to determine the 'expected' net income (NI) if ONLY the **selling price per unit** increases by 10%.

	BASE CASE R	**SENSITIVITY CASE R**
Sales	40 000	
Variable costs	20 000	
Fixed costs	12 000	
Depreciation	4 000	
EBIT	4 000	
Tax (27%)	1 080	
Earnings after tax	2 920	

1.2 Calculate the percentage change in net income if there is a 10% increase in the selling price per unit.

Solution

1.1

	BASE CASE R	**SENSITIVITY CASE R**
Sales	40 000	44 000
Variable costs	20 000	20 000
Fixed costs	12 000	12 000
Depreciation	4 000	4 000
EBIT	4 000	8 000
Tax (27%)	1 080	2 160
Earnings after tax	2 920	5 840

1.2

$\%\Delta = (NI_S - NI_B) / NI_B$
$= (5\ 840 - 2\ 920) / 2\ 920$
$= 100\%$

2. The MultiFirm Group is analysing a proposed project.

The group forecasts to sell 4 500 units at a selling price of R1 000 per unit in the worst case scenario and to sell 6 000 at a selling price of R1 500 per unit in the best case scenario.

The expected variable cost per unit is R600 at the lower bound and R800 at the upper bound. The expected fixed costs are R320 000 at the lower bound and R520 000 at the upper bound.

The depreciation expense is R250 000.

Complete the budgeted/pro forma statement of financial performance during the worst case scenario.

	WORST CASE **R**
Sales	
Variable costs	
Fixed costs	
Depreciation	
EBIT	
Tax (27%)	
EARNINGS AFTER TAX	

Solution

	WORST CASE **R**
Sales	4 500 000
Variable costs	3 600 000
Fixed costs	520 000
Depreciation	250 000
EBIT	130 000
Tax (27%)	35 100
EARNINGS AFTER TAX	94 900

PART FOUR: LONG-TERM FINANCIAL MANAGEMENT: FINANCING

CHAPTER 15: THE COST OF CAPITAL

INTRODUCTION

This chapter covers the separate parts of a firm's cost of capital by considering the cost of the different capital sources namely long-term debt, preference shares and ordinary shares. The cost of ordinary shares is covered by using both the constant growth model and the capital asset pricing model (CAPM). Thereafter the different parts are covered highlighting cost of capital in completeness through the weighted average cost of capital (WACC) calculations.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- accurately calculate the cost of long-term debt
- accurately calculate the cost of preference shares
- accurately calculate the cost of ordinary shares
- accurately calculate the weighted average cost of capital (WACC)
- use cost of capital for the purpose of investment decisions.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. Doorstep Clothing Ltd approaches Stable Bank for a long-term loan to purchase a new delivery vehicle. After considering all the risk criteria, Stable Bank grants the loan of R800 000, at a cost of 12.75%, which is prime plus 1. The marginal tax rate is 27%. The firm's cost of the loan or after-tax cost of long-term debt is

(a) 12.75%.
(b) 11.75%.
(c) 9.75%.
(d) 9.31%.
(e) 27.31%.

2. Doorstep Clothing Ltd considers issuing 1 000 000 12% preference shares, which are expected to sell at R3.50 each. Flotation costs are expected to amount to R1 per share. The firm's cost of preference shares is

(a) 12.75%.
(b) 12.00%.
(c) 11.00%.
(d) 15.50%.
(e) 16.80%.

3. Doorstep Clothing Ltd's ordinary shares are trading at R12.50 on the stock exchange. The firm expects to pay a dividend of 80 cents per share at the end of the coming year. The growth rate in dividends is a constant 11%. The cost of the firm's ordinary shares is

(a) 27.00%.
(b) 17.40%.
(c) 12.50%.
(d) 11.00%.
(e) The cost of ordinary share cannot be calculated.

4. Blue Lagoon Ltd has a beta coefficient of 0.9. The risk-free rate of return is 15.25% and the market rate of return is 23%. The firm's the cost of retained earnings is

(a) 23.00%.
(b) 22.23%.
(c) 15.25%.
(d) 15.23%.
(e) 7.75%.

5. How many of the following statements is/are *incorrect*?

   (i) The cost of capital cannot be used as a firm's discount rate for valuation purposes.
   (ii) The cost of capital is the link between long-term financing decisions.
   (iii) The cost of capital is the rate of return a firm must earn on its investments in order to maintain the market value of its ordinary shares.
   (iv) The cost of capital is the rate of return required by suppliers of capital in order to attract their funds to a firm.

(a) none
(b) one
(c) two
(d) three
(e) four

Answers to short questions

1. d
2. e
3. b
4. b
5. b

EXAMPLES OF OPEN QUESTIONS

1. Would companies in a country with a relatively high tax rate for companies tend to use mainly equity finance or mainly debt finance?

2. Strategists and investment analysts expect Bulk Gas Ltd to maintain a constant growth rate of 13% in the next few years. The firm makes use of the following optimal capital structure: Debt financing at 20%, Preference shares at 25% and Equity capital at 55%. The following information is also available:

The firm can obtain new long-term debt financing by issuing 15-year long-term debentures, at a discount of R40 each, on the capital market. The par value of each debenture is R1 000. They have a coupon interest rate of 13% p.a.

Bulk Gas Ltd intends to fund additional financing by issuing 10% preference shares with a par value of R60 and a dividend of R6 per share per year. The firm expects to realise R52 per share.

The firm can issue new ordinary shares by noting the following information: The dividend growth rate has been constant over the last few years at 13% and the firm's last dividend was paid at R3.20 per share. The present market price of the firm's shares is R45 and the floatation costs are estimated at R1.50 per share.

Calculate the firm's weighted average cost of capital. Work with 2 decimals <u>throughout</u> your calculations.

Solution

The weights or percentages (%s) of the various capital sources were given in this example.

The various component costs are:

Before-tax cost of debt
PV = R960; FV = -1 000; PMT = -130; n = 15; YTM = 13.64%

After-tax cost of debt
k_i = k_d x (1 - t)
k_i = 13.64 (1 – 0.27) = 9.96%

Cost of preference shares
k_p = D_p / N_p
= 6 / 52 = 11.54%

Cost of new ordinary shares
k_n = (D_1 / N_n) + g

First calculate D_1 and the net proceed per share:
D_0 was given as RR3.20
D_1 = D0 (1 + g)
= 3.20 (1 + 0.13)
= R3.62
Net proceed per share = 45 – 1.50 = R43.50

k_n = $(D_1 / N_n) + g$
= (3.62 / 43.50) + 0.13
= 21.32%

WACC = Sum of [weight of EACH capital source x cost of EACH capital source]
OR
= $[D/V) \times k_i] + [(P/V) \times k_p + [(E/V) \times k_n]$

Capital source	**Weight**	**Cost**	**Weight x Cost**
Debt	0.20	9.96	1.99
Preference shares	0.25	11.54	2.89
Ordinary shares	0.55	21.32	11.73
		Sum	**WACC = 16.61%**

PART FOUR: LONG-TERM FINANCIAL MANAGEMENT: FINANCING

CHAPTER 16: LEVERAGE AND CAPITAL STRUCTURE

INTRODUCTION

This chapter covers the behaviour of costs if there is a change in production quantity as well as the interrelationships between costs and sales. This is also highlighted when conducting a breakeven analysis. Firms are financed with either equity or debt or with a combination of both (referred to as the capital structure of a firm); therefore, firms are exposed to various types of leverage (operating, financial and total). The optimal capital structure (way of financing a firm) is important and the considerations to be taken into account when determining the optimal capital structure are addressed.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- classify cost as fixed, variable or semi-variable
- accurately calculate the breakeven point of a firm by means of breakeven analysis
- evaluate the influence of operating and financial leverage on a firm
- determine the influence of financial leverage on the earnings, optimal capital structure and value of a firm
- evaluate the appropriateness of financial leverage in view of practical considerations.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. Use the following information to calculate the breakeven volume and breakeven value.

Selling price per unit	R250	Variable cost per unit	R100
Expected quantity	50 000 units	Total fixed cost	R600 000

(a) 1 715 units; R428 75012
(b) 1 800 units; R450 000
(c) 2 400 units; R600 000
(d) 4 000 units; R1 000 000
(e) 6 000 units; R1 500 000

Use the following information to answer multiple choice questions 2 to 4.

The applicable tax rate is 27% and the interest on debt is 12% (debt amounts to R75 000). If there is an increase of 20% in sales, EBIT will change from R200 000 to R300 000. Earnings after tax will change from R139 430 to R212 430 if there is a 20% increase in sales.

2. Calculate the degree of operating leverage (DOL) if there is a 20% increase in sales.

(a) 0.20 times
(b) 0.50 times
(c) 1.00 times
(d) 2.50 times
(e) 5.00 times

3. Calculate the degree of financial leverage (DFL) if there is a 20% increase in sales.

(a) 0.50 times
(b) 0.95 times
(c) 1.00 times
(d) 1.05 times
(e) 2.50 times

4. Calculate the degree of total leverage (DTL) if there is a 20% increase in sales.

(a) 1.00 times
(b) 2.38 times
(c) 2.50 times
(d) 2.63 times
(e) 3.75 times

5. __________ is the firm's ability to adjust its sources of funds upwards or downwards in response to major changes in the need for funds.

(a) Flexibility
(b) Timing
(c) Control
(d) Planning
(e) Sustainability

Answers to short questions

1. d
2. d
3. d
4. d
5. a

EXAMPLES OF OPEN QUESTIONS

1. The Banana Company is comparing two different capital structures, namely an all-equity plan (Plan 1) and a levered plan (Plan 2). Under Plan 1, the company would have 150 000 shares in issue. Under Plan 2 there would be 60 000 shares in issue and R1.5 million in debt. The interest rate on the debt is 11%. Ignore taxes.

1.1 If EBIT is R200 000, which plan will result in the higher EPS?

1.2 If EBIT is R700 000, which plan will result in the higher EPS?

1.3 Calculate the break-even EBIT.

Solution

1.1

	Plan 1	Plan 2
EBIT	R200 000	R200 000
Interest	-	165 000
Earnings	R200 000	R 35 000
EPS	R1.33	R0.58

Plan 1 has the higher EPS when EBIT is R200 000.

1.2

	Plan 1	Plan 2
EBIT	R700 000	R700 000
Interest	-	165 000
Earnings	R700 000	R 535 000
EPS	R4.67	R8.92

Plan 2 has the higher EPS when EBIT is R700 000.

1.3 We need to find the EBIT where the EPS for Plan 1 and Plan 2 are equal. To calculate the EPS for Plan 1, we should calculate EBIT by the number of shares in issue. For Plan 2, we will follow the same route, except that we will now have to deal with the interest payable on the debt. Thus, the interest payable will reduce EBIT, which we need to divide by the number of shares in issue.

EBIT / Number of shares in issue = (EBIT – interest payable) / Number of shares in issue

EBIT / 150 000	=	[EBIT – 0.11(R1 500 000)] / 60 000
60 000(EBIT)	=	150 000(EBIT) – 24 750 000 000
EBIT	=	R275 000

2. Mr Goldfinger is the financial manager of a listed company. He is analysing the firm's capital structure and has asked you to assist him in the analysis. The following information on the CURRENT SITUATION is available:

Sales (Q)	10 000 units
Sales price per unit (S)	R10
Variable cost per unit (VC)	R5
Fixed cost (FC)	R10 000
Long term debt outstanding	R150 000
Interest rate on all long term debt (R_D)	5%
Company tax rate (T_C)	30%

Number of ordinary shares in issue	10 000
Dividend pay-out ratio	100% (all earnings paid out)
Growth rate in earnings and dividends (g)	0 (all earnings are paid out)
Beta coefficient	1.50
Risk-free rate of return	5%
Market return	9%

Mr Goldfinger conducted a capital budgeting exercise and found that two potential capital projects are acceptable in terms of his stated internal rate of return criterion. He needs **R 100 000** additional (external) funds to finance the projects.

There are 3 possible ways in which to finance it:

- **Scenario 1**: with 100% debt
- **Scenario 2**: with 100% equity (in this case 10 000 new shares will be issued)
- **Scenario 3**: with 50% debt and 50% equity

The probability that sales prices will increase is as follows:

Probability	0.3	0.4	0.3
Sale price per unit (S)	R12	R14	R16

The number of units produced (Q) variable (VC) and fixed (FC) costs remain unchanged.

2.1 Construct the CURRENT statement of financial performance.
2.2 Calculate the current EPS and share price.
2.3 Calculate the expected EPS and share prices in each of the three financing scenarios.
2.4 How should the projects be financed?

Solution

2.1 Current statement of financial performance.

	R	
Sales	100 000	(S*Q)
Variable cost	50 000	(VC*Q)
Fixed cost	10 000	
EBIT	**40 000**	
Interest	7 500	(5% of R150 000)
EBT	32 500	
Tax = 30%	9 750	
Earnings after tax	**22 750**	

2.2 Current EPS and share price

EPS	**2.28**	(Net income / 10 000 shares)
Required rate of return (R)	**11%**	$E(R_i) = r_f + \beta\ [E(r_M) - r_f] = 5 + 1.5(9 - 5)$
Growth rate	**0%**	(All earnings are paid out as dividends)
DPS = EPS = D_0	**2.28**	
Share price (P_0)	**R20.73**	$D_1 / (r - g)$ where g = 0; $P_0 = 2.28 / 0.11$ Refer to Chapter 7

2.3 EPS and share price in each scenario

Scenario 1: Finance with 100% Debt

Probability	0.3	0.4	0.3	
Sales	120 000	140 000	160 000	(S*Q)
Variable cost	50 000	50 000	50 000	(VC*Q)
Fixed cost	10 000	10 000	10 000	
EBIT	**60 000**	**80 000**	**100 000**	
Interest	12 500	12 500	12 500	= 0.05 × old debt + 0.05 × new debt = 0.05 × 150 000 + 0.05 × 100 000
EBT	47 500	67 500	87 500	
Tax (30%)	14 250	20 250	26 250	
Earnings after tax	**33 250**	**47 250**	**61 250**	
EPS	**3.33**	**4.73**	**6.13**	= Net income / (old nr of shares) = Net income / 10 000

Expected (average) EPS	4.73	$= (0.3 \times 3.33) + (0.4 \times 4.73) + (0.3 \times 6.13)$
Standard deviation	1.08	$= \{[0.3 \times (3.33 - 4.73)^2] + [0.4 \times (4.73 - 4.73)^2] + [0.3 \times (6.13 - 4.73)^2]\}^{0.5}$
Required rate of return	11%	
Growth rate	0%	
DPS= exp. EPS= D_0	R4.73	
Share price (P_0)	**R43.00**	$D_1 / (r - g)$ where g

Scenario 2: Finance with 100% equity – Zero debt used

Probability	0.3	0.4	0.3	
Sales	120 000	140 000	160 000	
Variable cost	50 000	50 000	50 000	
Fixed cost	10 000	10 000	10 000	
EBIT	**60 000**	**80 000**	**100 000**	
Interest	7 500	7 500	7 500	= 5% of old debt = 0.05 x 150 000
EBT	52 500	72 500	92 500	
Tax (30%)	15 750	21 750	27 750	
Earnings after tax	**36 750**	**50 750**	**64 750**	
EPS	**1.84**	**2.54**	**3.24**	= Net income / (old + new nr of shares) = Net income / (10 000 + 10 000)

Expected (average) EPS	2.54	= (0.3 × 1.84) + (0.4 × 2.54) + (0.3 × 3.24)
Standard deviation	0.54	$= \{[0.3 \times (1.84 - 2.54)^2] + [0.4 \times (2.54 - 2.54)^2] + [0.3 \times (3.24 - 2.54)^2]\}^{0.5}$
Required rate of return	11%	
Growth rate	0%	
DPS = exp. EPS = D_0	R2.54	
Share price (P_0)	**R23.09**	

Scenario 3: Finance with 50% debt & 50% equity

Probability	0.3	0.4	0.3	
Sales	120 000	140 000	160 000	(S*Q)
Variable cost	50 000	50 000	50000	(VC*Q)
Fixed cost	10 000	10 000	10 000	
EBIT	**60 000**	**80 000**	**100 000**	
Interest	10 000	10 000	10 000	= 0.05 × "old" debt + 0.05 × "new" debt = 0.05 × 150 000 + 0.05 × 50 000
EBT	50 000	70 000	90 000	
Tax (30%)	15 000	21 000	27 000	
Earnings after tax	**35 000**	**49 000**	**63 000**	
EPS	**2.33**	**3.27**	**4.2**	= Net income / (old + new nr of shares) = Net income / (10 000 + 5 000)

Expected (average) EPS	3.27	= (0.3 × 2.33) + (0.4 × 3.27) + (0.3 × 4.2)
Standard deviation	0.72	$= \{[0.3 \times (2.33 - 3.27)^2] + [0.4 \times (3.27 - 3.27)^2] + [0.3 \times (4.2 - 3.27)^2]\}^{0.5}$
Required rate of return	11%	
Growth rate	0%	
DPS = exp. EPS = D_0	R3.27	
Share price (P_0)	**R29.73**	

PART FOUR: LONG-TERM FINANCIAL MANAGEMENT: FINANCING

CHAPTER 18: DIVIDEND POLICY

INTRODUCTION

This chapter focuses on the influence of firms' dividend policies, payouts and other factors on their share prices. Various forms of dividends are discussed. The chapter also highlights the influence of share splits and repurchases.

LEARNING OUTCOMES

After studying this chapter, you should be able to:

- distinguish between the various forms of dividends
- describe the cash dividend payment processes
- illustrate the influence of cash dividends, share dividends and share splits on the value of the firm
- evaluate the factors that influence dividend policy
- apply a residual dividend policy
- debate whether dividends have an influence on a firm's share price.

Use the lecture discussions to cover each of the above learning outcomes and ensure that you understand and can apply all of the topics and content covered. Use the lecture slides only as an aid.

EXAMPLES OF SHORT QUESTIONS

1. The date before which a new purchaser of a share is entitled to receive a declared dividend, but on or after which he/she does not receive the dividend, is called the _________ date.

(a) ex-rights
(b) ex-dividend
(c) record
(d) payment
(e) declaration

2. A share is trading at R25.80. The firm has declared a dividend of R4.40. At what price will the share commence trading on the ex-dividend date?

(a) The price on the ex-dividend date cannot be determined
(b) R4.40
(c) R21.40
(d) R25.80
(e) R30.20

3. All else equal, a share split will lead to ______ in a shareholder's number of shares, a firm's liquidity will __________ and the earnings per share will __________ .

(a) a change; improve; decrease
(b) a change; decrease; decrease
(c) a change; improve; increase
(d) no change; improve; decrease
(e) no change; improve; increase

4. A firm has a return on equity (ROE) of 27.75%, a return on assets (ROA) of 28.50%. It also has a retention ratio of 40%, and therefore a dividend payout ratio of 60%. The firm's sustainable growth (g) rate is

(a) 28.50%.
(b) 27.75%.
(c) 16.65%.
(d) 11.40%.
(e) 11.10%.

5. A firm which adopts a residual dividend policy

(a) prefers to offer new securities for sale on a routine basis.
(b) prefers constant dividends to a constant debt-equity ratio.
(c) places a higher priority on funding its investment needs than on paying dividends.
(d) will pay regular cash dividends that are constant in amount.
(e) tends to also have a high dividend policy.

Answers to short questions

1. b
2. c
3. a
4. e
5. c

EXAMPLES OF OPEN QUESTIONS

1. ABC Manufacturers declared a R3.50 per share dividend on 10 May 2025. The last date to trade was 24 May 2025 while the ex-dividend date was 27 May 2025. The record date was 30 May 2025, and the dividend payment date was 4 June 2025.

(a) On what date did ABC Manufacturers close its share transfer book and produce a list of shareholders who would receive the declared dividend payment?
(b) If the share was traded on 13 May 2025, who would receive the dividend declared on 10 May 2025?

2. The BA Yacht Company's optimal capital structure calls for 40% debt and 60% equity financing. The firm calculated its weighted average cost of capital (WACC) of 12.70%.

The firm has the following investment opportunities:

Project	Cost	IRR
A	7 500 000	23%
B	7 500 000	13%
C	7 500 000	9%

The firm expects to have a earnings after tax (net profit after tax) of R10.5 million. If the firm bases its dividends on the residual policy, what will its payout ratio be?

Solution

Step 1: Calculate the WACC
The WACC was given as: WACC = 12.70%

Step 2: Determine the optimal capital budget
Projects A and B are acceptable since their IRRs exceed the cost of capital (WACC = 12.70%). As each of these two projects cost R7.5 million, the total capital budget is R15 million.

Step 3: Determine how much equity is necessary to finance the acceptable projects
60% of the total financing should be equity, thus 60% of R15 million = R9 million.

Step 4: Determine the residual dividend
Residual dividends = Earnings after tax – Equity required
= 10 500 000 – 9 000 000
= R1 500 000

Step 5: Determine the payout ratio
Payout ratio = Dividend / Earnings after tax
= 1 500 000 / 10 500 000
= 14.29%

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